



GE Medical Systems

Technical Publications

Direction 5133317

Revision 7

Signa® EXCITE 3.0T System Upgrade from 11.x to 12.x

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003



GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1 ..	Sep 27, 2004 ..	Initial Release.
2 ...	May 5, 2005 ..	Sec 2 – Added “saveINFO” and ACGD to HFD–S Upgrade procedure. Sec 4 – Updated the “added” cable maps by adding optional remote MRU module, cables, and connections and HFD–S/PDU cabinet upgrade. Sec 5 – Corrected sub–section title. Sec 6 – Added instructions for removing gradient cable foam and installing clamp blocks (MRIhc06323). Changed Run 1206 to 1220 and added cable connection instructions. Sec 7 – Added jumper board part number. Sec 8 – Added step for verifying network switch. Added optional location for RF Splitter box (MRIhc06149).
3 ...	Jun 8, 2005 ..	Sec 8 – Added step for moving 50 Ohm terminator from Mux1 to Mux2 J6 on RRF module.
4 ...	Dec 6, 2005 ..	Sec 4 – Revised paragraph on cable information. Corrected GOC Power Strip connection designators. Sec 8 – Replaced photo of rear of RRF Module with one showing correct cable routing.
5 ..	Aug 18, 2006 ..	Sec 2 – Added “Power On” and “Host Monitor Setup and Cal” procedures. Sec 8 – Revised the instructions for installing new boards in RRF Chassis and added the alternative RF–DIF board, 5250028.
6 ...	Mar 7, 2007 ..	Section 8: Updated instructions for installing 8 Channel upgrade. Revised 16 Channel installation procedures by updating MUX board part number and note for recommendation on existing MUX board. Added instructions for storing Service Cable Kit.
7 ..	Jun 17, 2009 ..	Section 6: Added 6–3–10 for installing laser warning labels (PQR 13246299).

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	7	2–2	2	TAB 5		TAB 7		8–7 to 8–11	6
Damage in Trans.	–	2–3	1	PENETRATION PANEL		OPERATOR		TAB 9	
Direction 2128126 ...	1*	2–4	2	2164904–5.....	–	WORKSPACE		RF ACCESSORY	
A	7	2–5	5	5–1 to 5–2	2	2164904–7.....	–	CABINET	
i	1					7–1 to 7–2	2	2164904–9.....	–
TAB 1		TAB 3		TAB 6		TAB 8		9–1 to 9–2	1
GETTING STARTED		POWER OFF		MAGNET ROOM		SYSTEM CABINET		TAB 10	
2164904–1.....	–	2164904–3.....	–	2164904–6.....	–	2164904–8.....	–	COMPLETION	
1–1 to 1–6	1	3–1	1	6–1	7	8–1 to 8–4	2	2164904–11	–
TAB 2		TAB 4		6–2 to 6–18	2	8–5	6	10–1 to 10–2	1
UPGRADE		SYSTEM UPGRADE		6–19 to 6–20	7	8–6	2		
INSTALLATION		CABLES							
PROCESS		2164904–4.....	–						
2164904–2.....	–	4–1 to 4–10	4						
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SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 3.0T EXCITE (Release G3, EXCITE II) to Signa 3.0T EXCITE HD (Release 12.x, EXCITE III). Refer to illustration 1-1 for pictorial overview of upgraded system.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 3.0T EXCITE HD system are listed in the *Signa 3.0T EXCITE Upgrade Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 3.0T EXCITE HD Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMLs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. The IIS key will need to be re-generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE HD (Release 12.x) upgrade to GoldSeal which includes:

- Driver Module
- SRI 2
- 8 Channel Switch
- LPCA
- Rear Bridge
- Cable Take-Up
- Exciter / Receiver Boards
- Front Bridge

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION**1-7-5-1 Sites in Americas**

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-7-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997

Fax: (414)–747–6855

Field ASPEN: 1–800–525–1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL

Quai 45 à 49 – 91090 LISSES – FRANC

Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41

Fax: (+33) 1–60–86–12–31

For non-Gold Seal returns in Europe:

European Renewable Resources Center

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2

Quai 45 à 49 – 91090 LISSES – FRANCE

TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

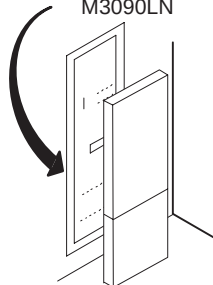
Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

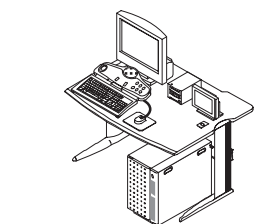
1-10 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment replaced and modified by Signa 3.0T EXCITE 11.x to 12.x Upgrade.

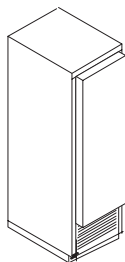
MODIFIED BY
EXCITE UPGRADE KIT
M3090LN



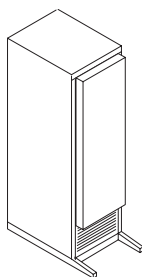
PENETRATION
PANEL
(PP1)



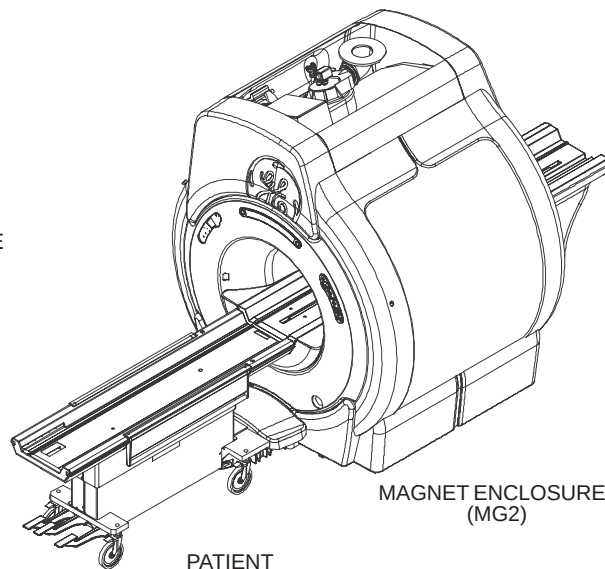
OPERATOR WORKSPACE
(OW1)



SYSTEM CABINET
(MR2)



RF ACCESSORY
CABINET
(MR1)



MAGNET ENCLOSURE
(MG2)

PATIENT
TRANSPORT
(PT1)

SIGNA 3T EXISTING HARDWARE REPLACED OR MODIFIED UPGRADE TO EXCITE HD (Release 12.x)

ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

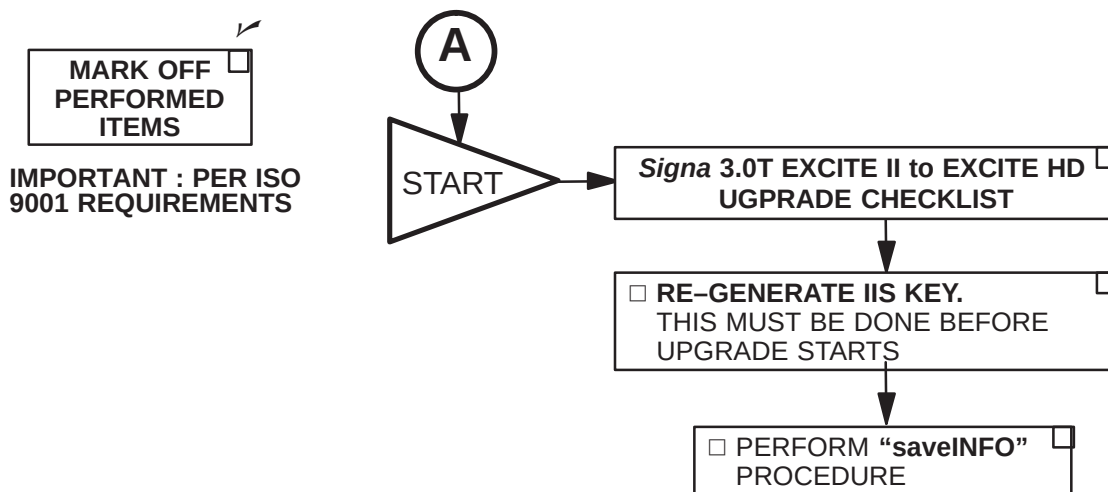
2-2 UPGRADE PROCEDURE

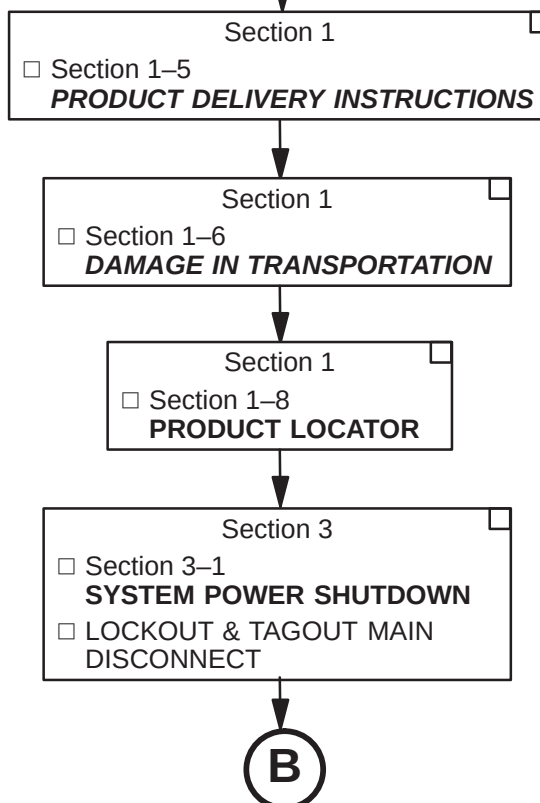
The following upgrade flow charts contain information for upgrading to Signa EXCITE 1.5T Release 12.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa EXCITE equipment has been delivered together. **Make sure that every part required is available before starting.**

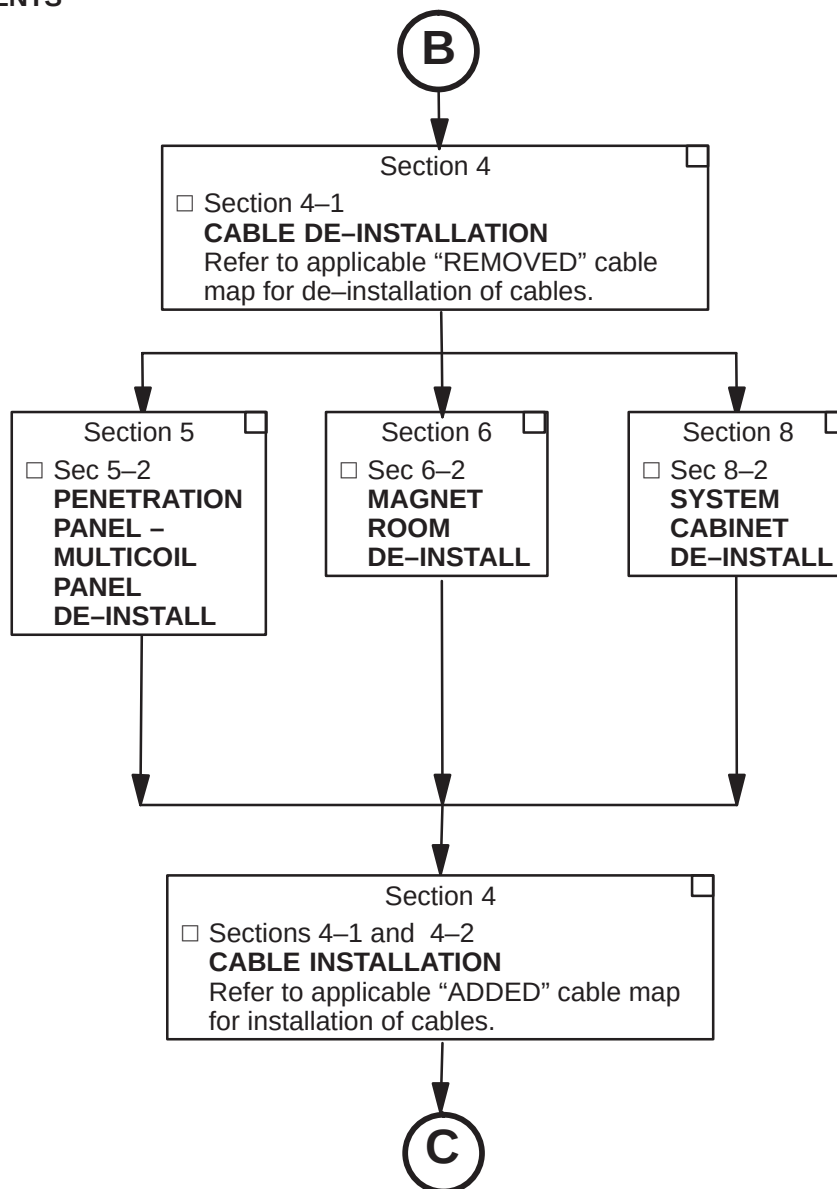
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

 Procedures to be Completed by Field Engineer


 Procedures to be Completed by FE and/or Mechanical Installers


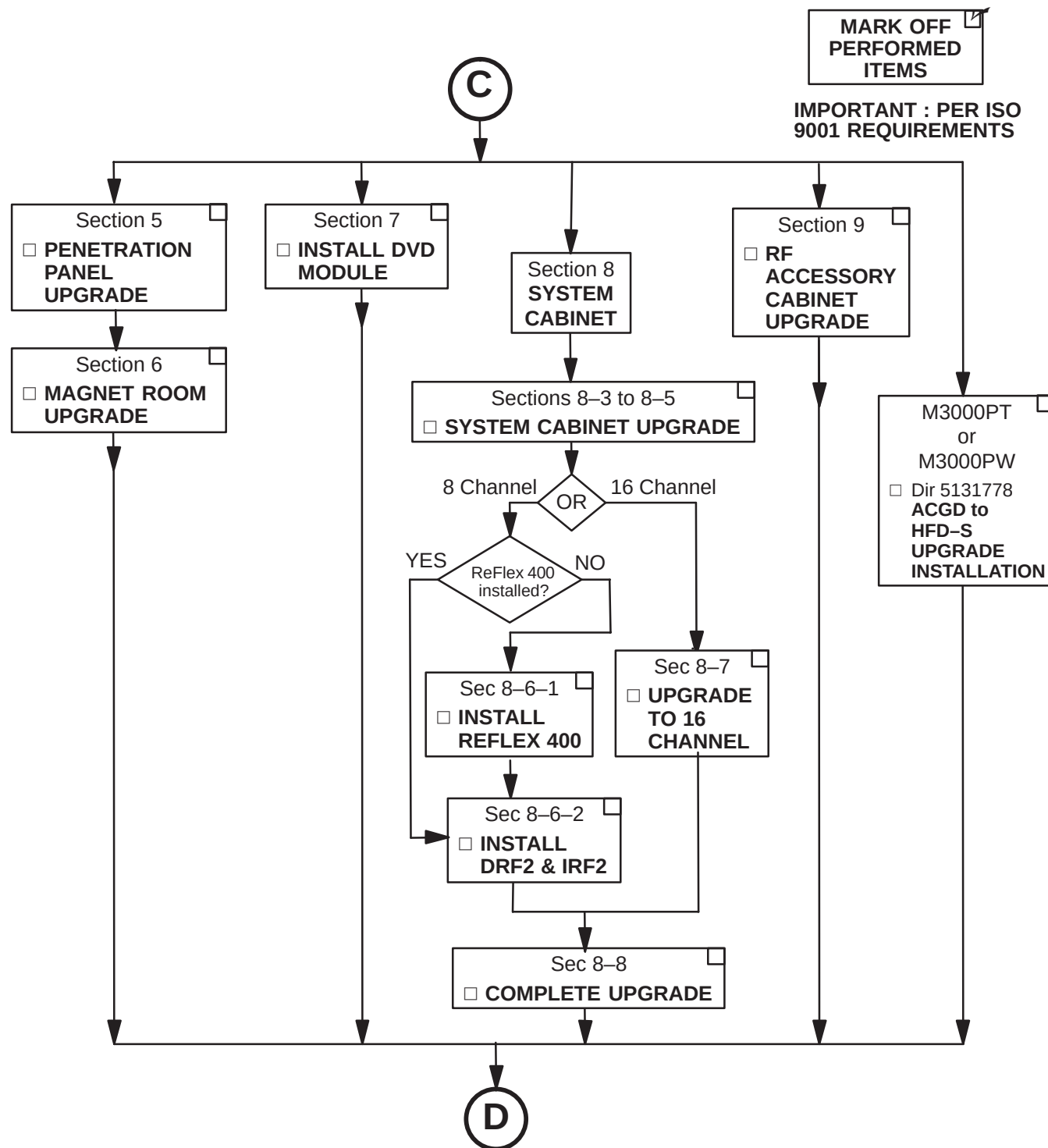
De-Installation Procedures to be Completed by FE and/or Mechanical Installers

**IMPORTANT : PER ISO
9001 REQUIREMENTS**



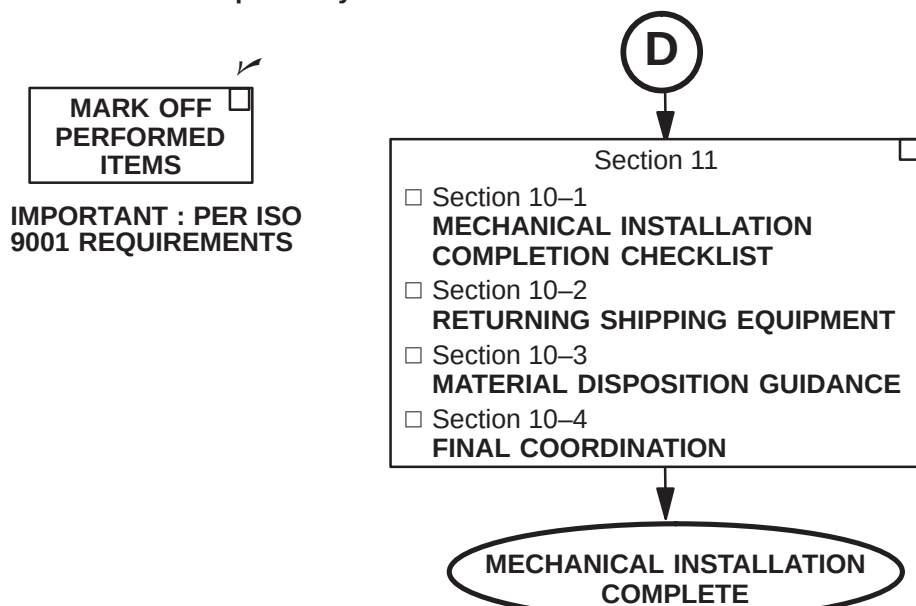
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

 Procedures to be Completed by FE and/or Mechanical Installers

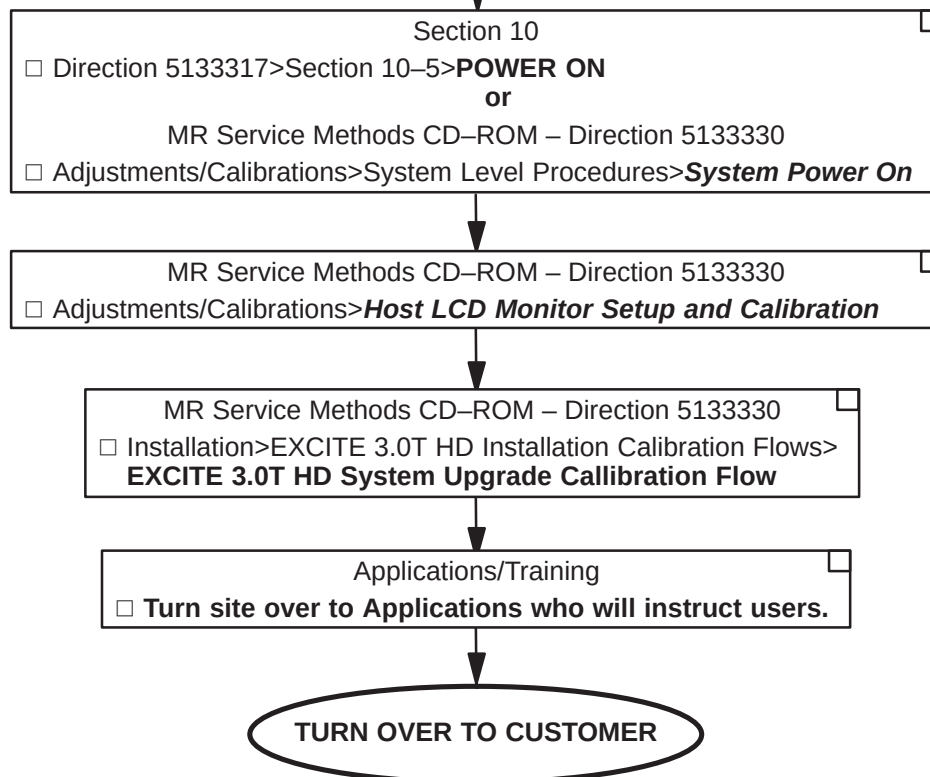


2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Make sure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM UPGRADE CABLES

4-1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading to Signa EXCITE 3.0T Release 12.x.

A double set of “Removed” and “Added” cables maps is provided. One set for the equipment room and the other for the magnet room. Tape maps in a convenient location to check off affected “Removed” and “Added” cables.

The Affected “Removed” Cable maps are located on the following pages.

- Page 4-3 Affected (removed) Cables for Upgrading to Signa 3.0T EXCITE Release 12.x (Equip. Room)
- Page 4-5 Affected (removed) Cables for Upgrading to Signa 3.0T EXCITE Release 12.x (Magnet Room)

The Affected “Added” Cable maps are located on the following pages.

- Page 4-7 Affected (added) Cables for Upgrading to Signa 3.0T EXCITE Release 12.x (Equip. Room)
- Page 4-9 Affected (added) Cables for Upgrading to Signa 3.0T EXCITE Release 12.x (Magnet Room)

Upon completion of connecting cables, return fold-out map to the binder for future reference.

4-2 SORTING AND ROUTING CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.

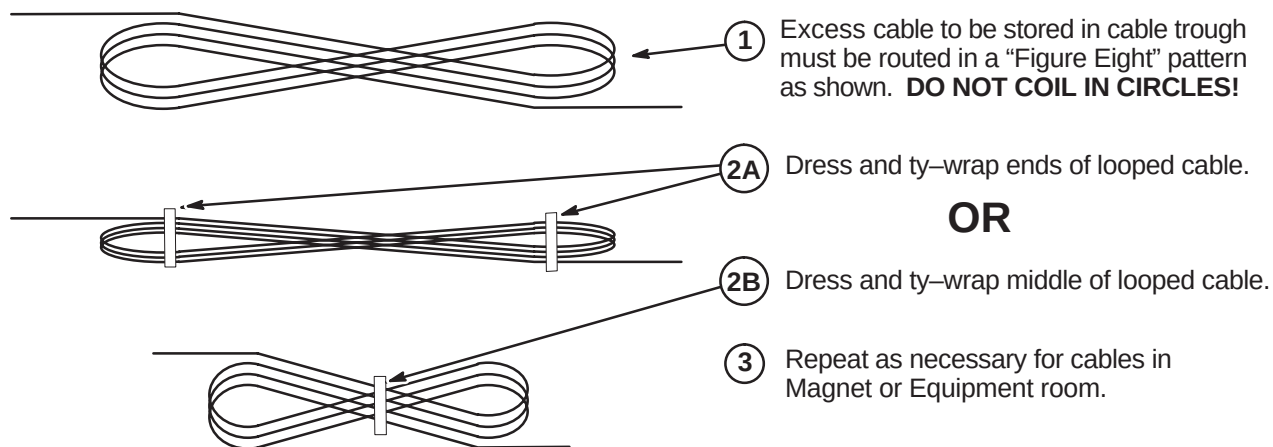
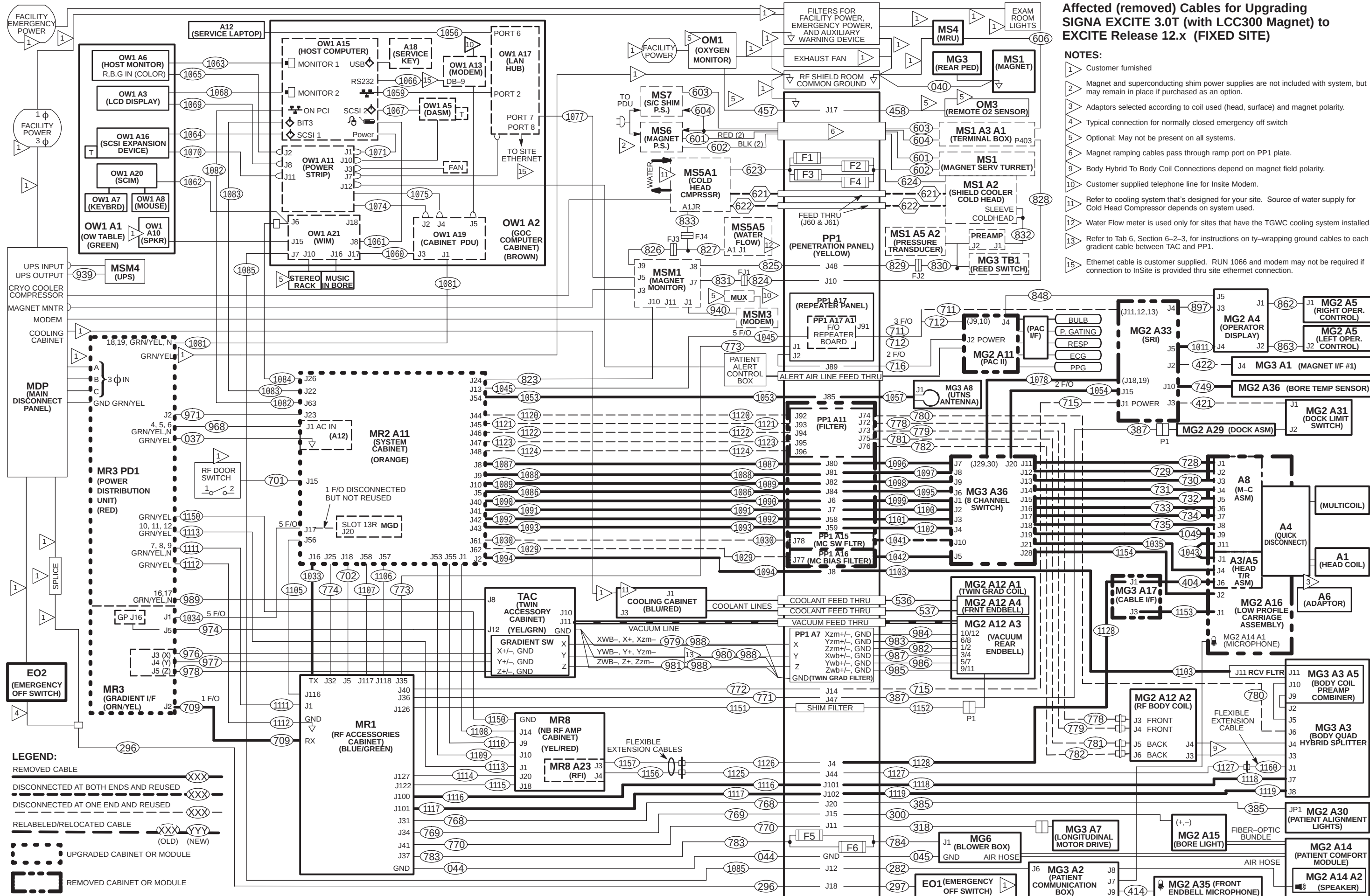
4-2 SORTING AND ROUTING CABLES (Continued)**PROPER STORAGE OF EXCESS CABLES**

ILLUSTRATION 4-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.
5. Rear Pedestal Cables – It is desired to keep the gradient cables separated from the signal and bias cables.

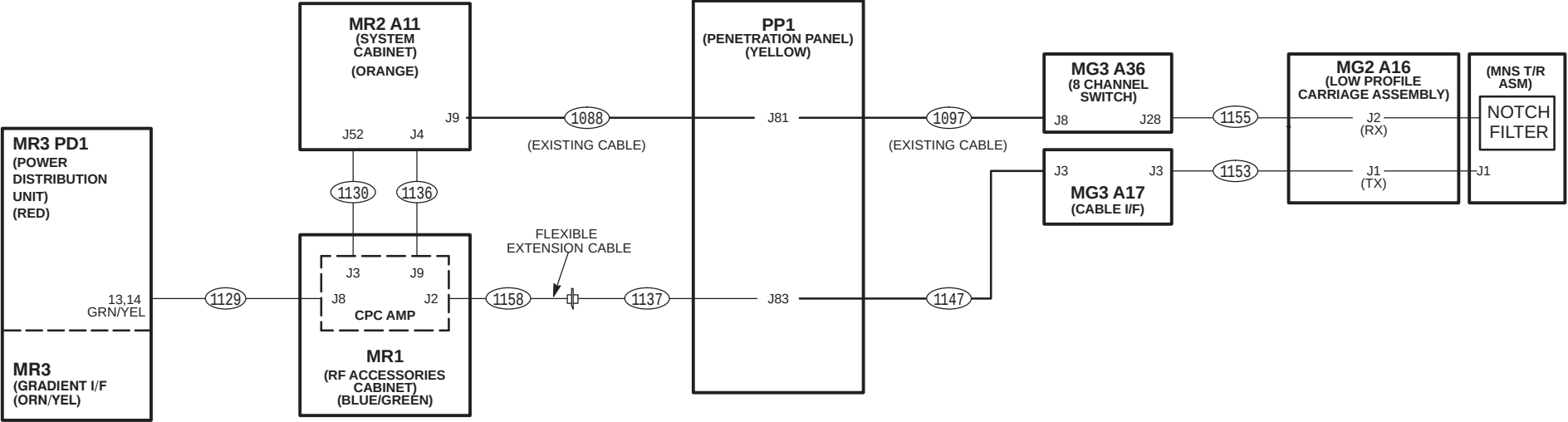


Affected (removed) Cables for Upgrading SIGNA EXCITE 3.0T (with LCC300 Magnet) to EXCITE Release 12.x (FIXED SITE)

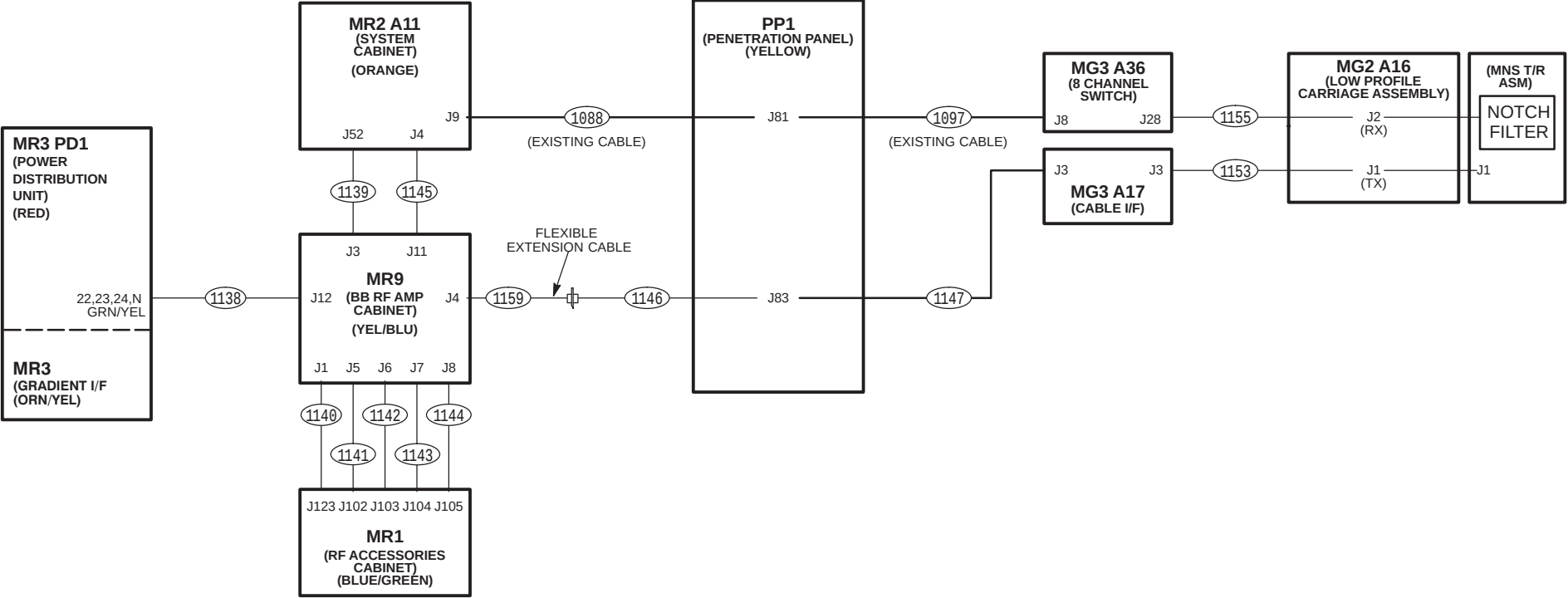
NOTES:

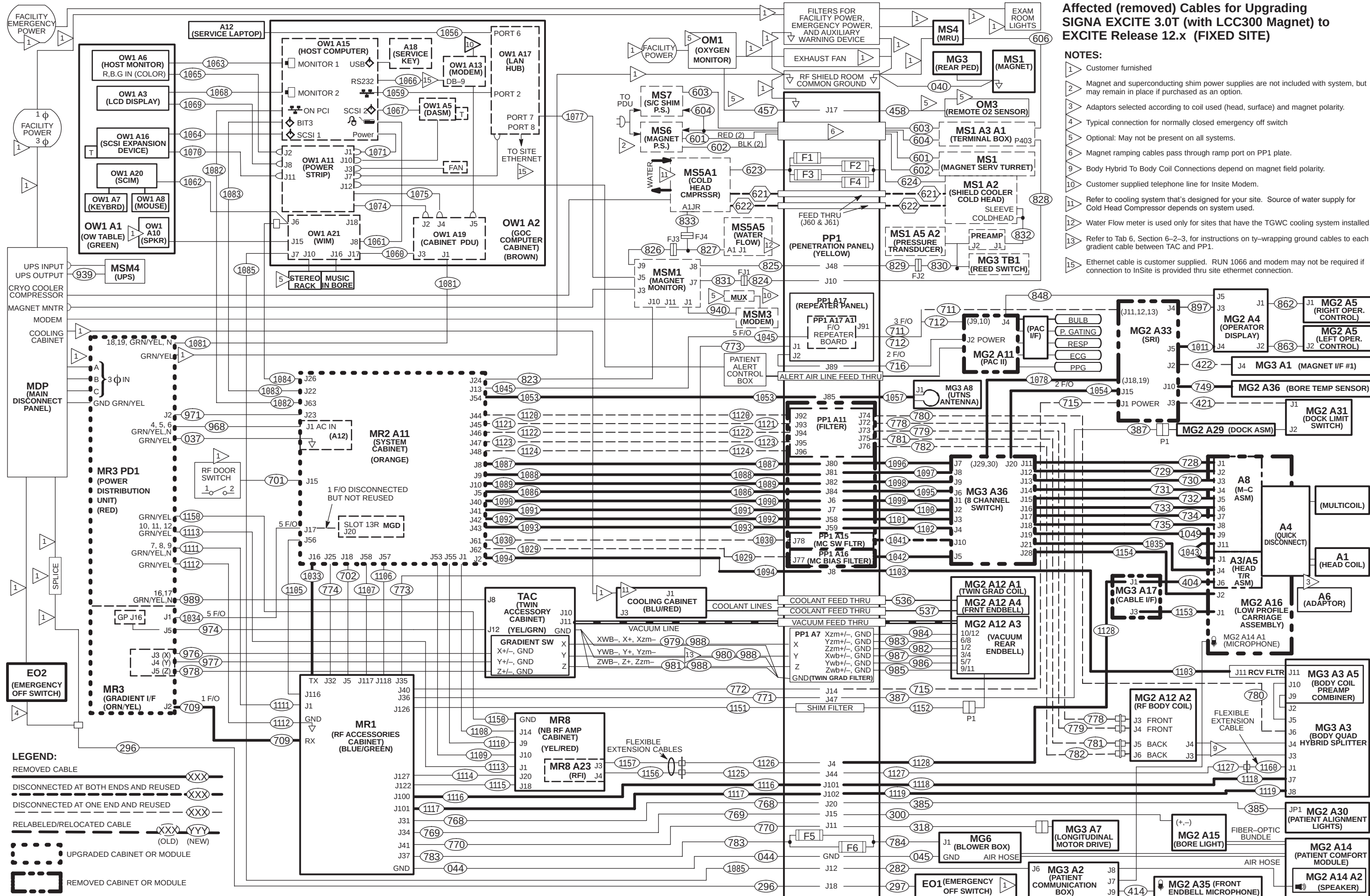
- Customer furnished
- Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- Adaptors selected according to coil used (head, surface) and magnet polarity.
- Typical connection for normally closed emergency off switch
- Optional: May not be present on all systems.
- Magnet ramping cables pass through ramp port on PP1 plate.
- Body Hybrid To Body Coil Connections depend on magnet field polarity.
- Customer supplied telephone line for Insite Modem.
- Refer to cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used.
- Water Flow meter is used only for sites that have the TGWC cooling system installed.
- Refer to Tab 6, Section 6-2-3, for instructions on ty-wrapping ground cables to each gradient cable between TAC and PP1.
- Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

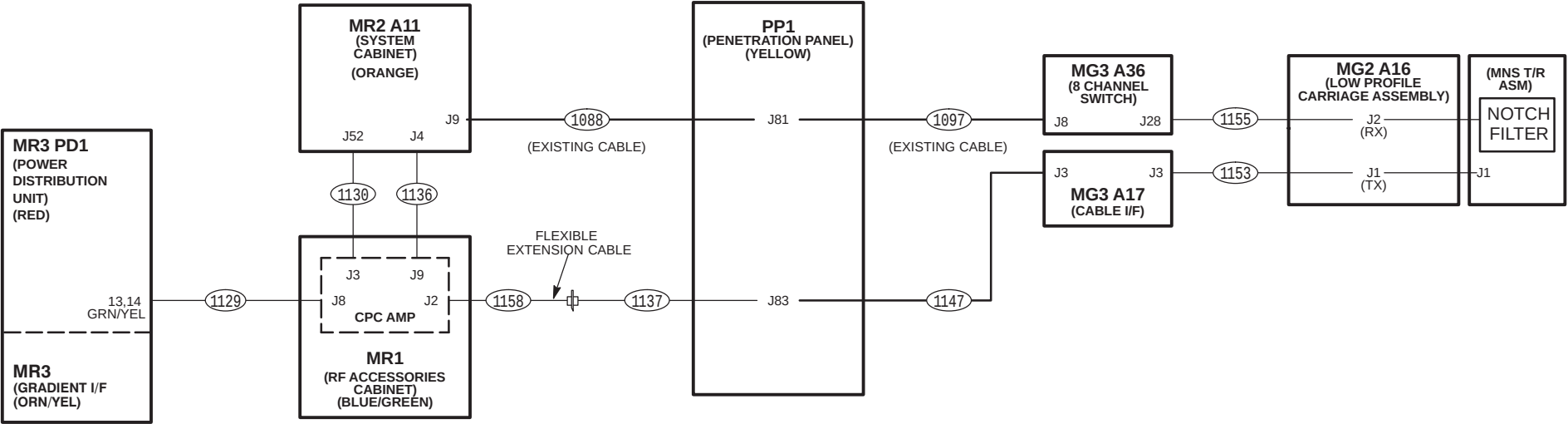


SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

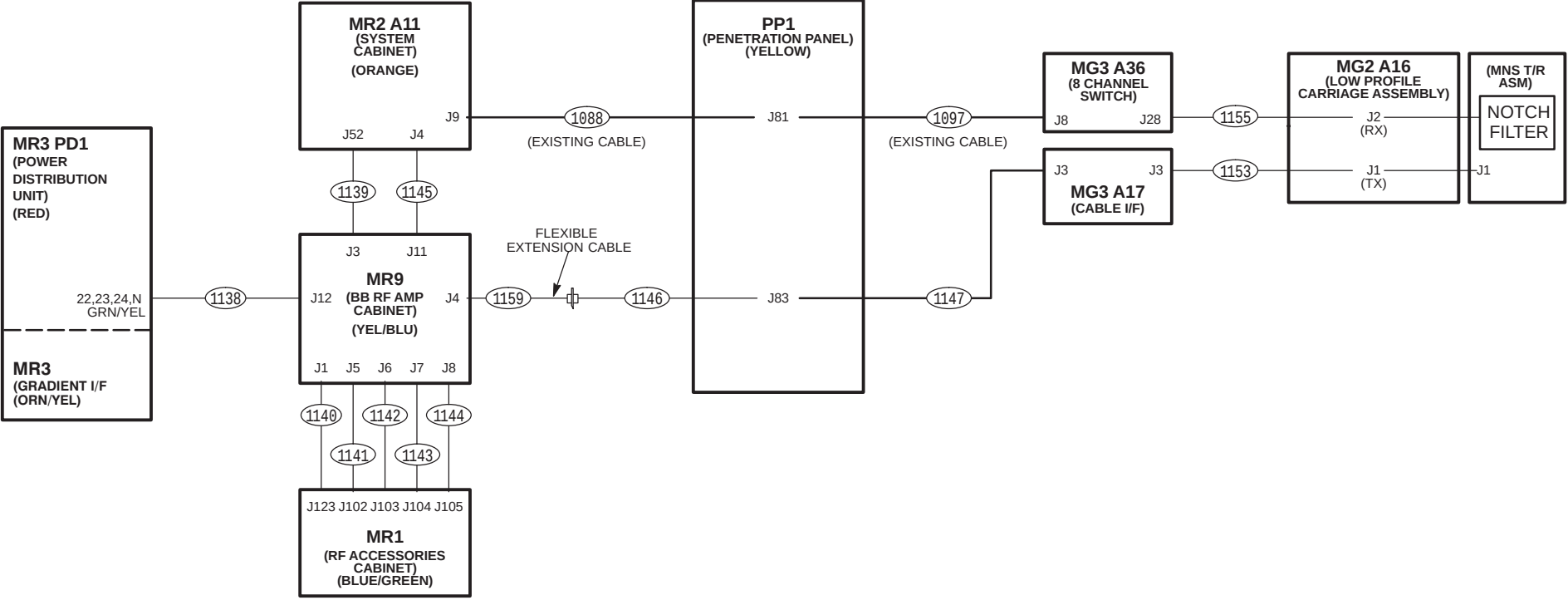




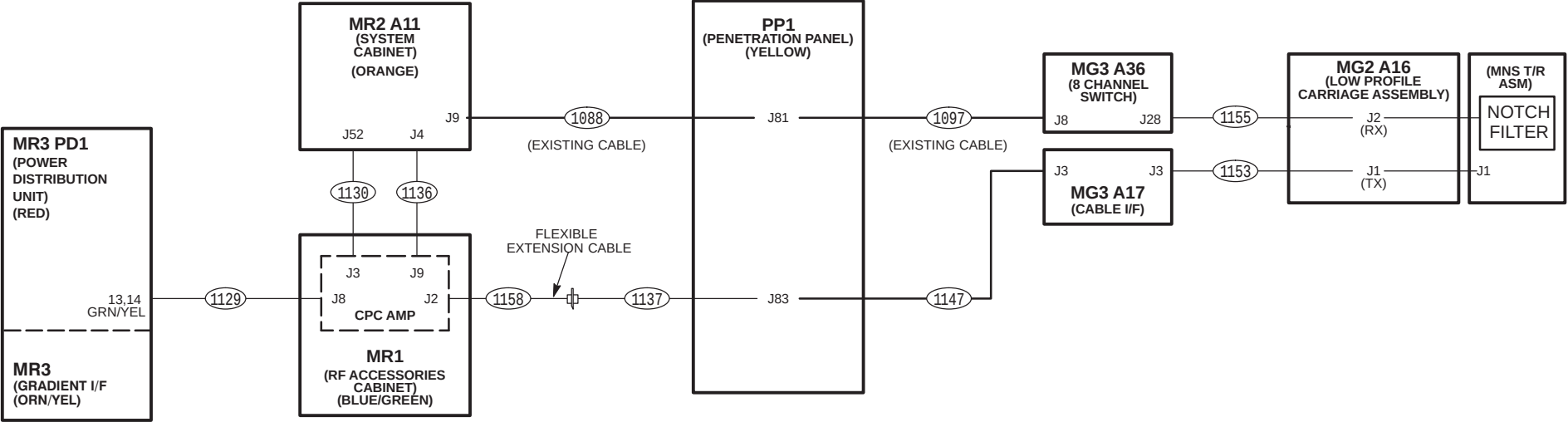
SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



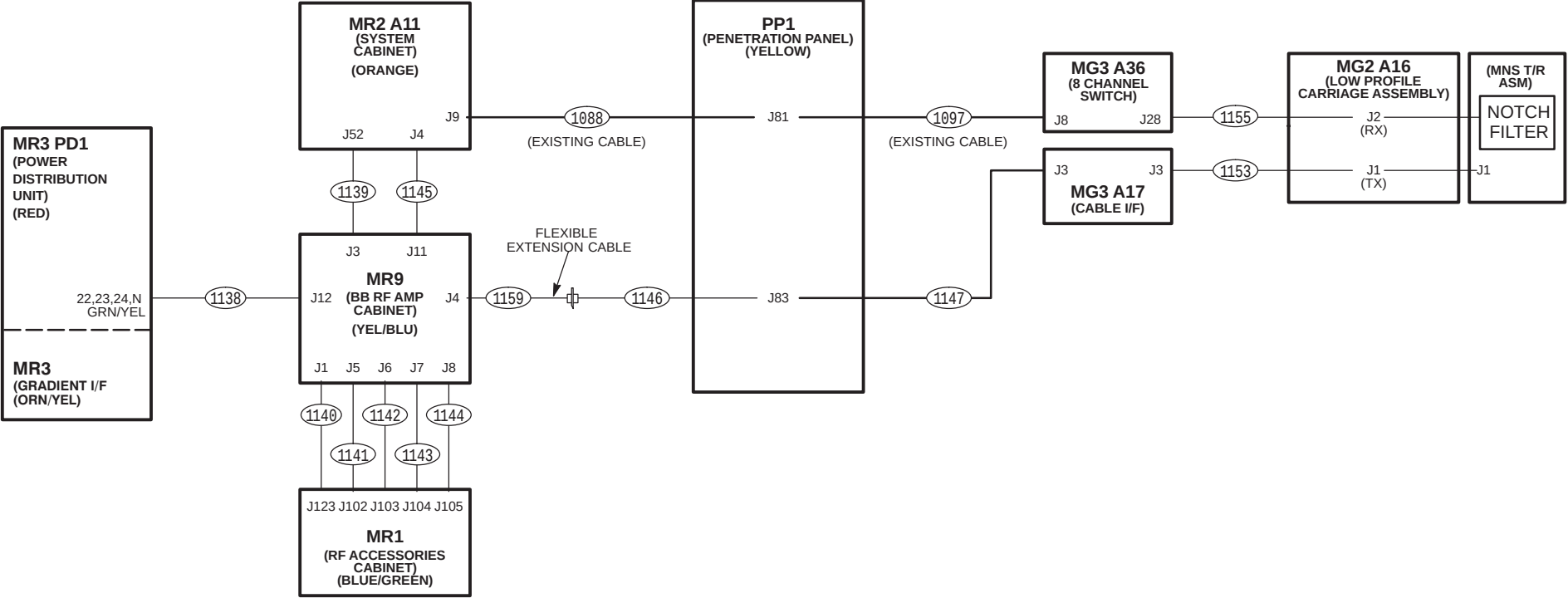
SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



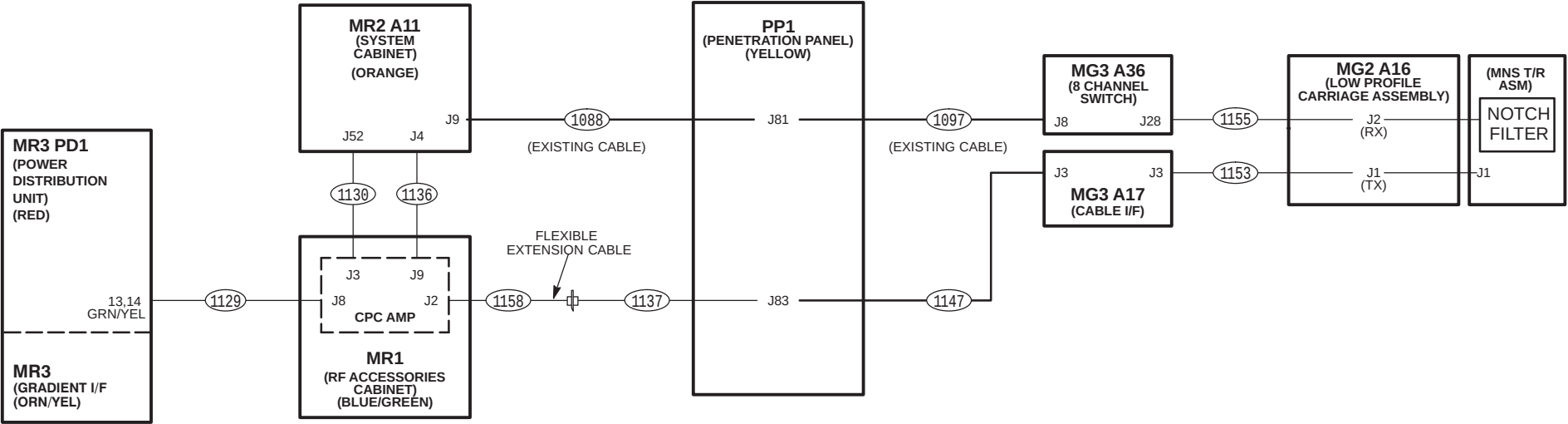
SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



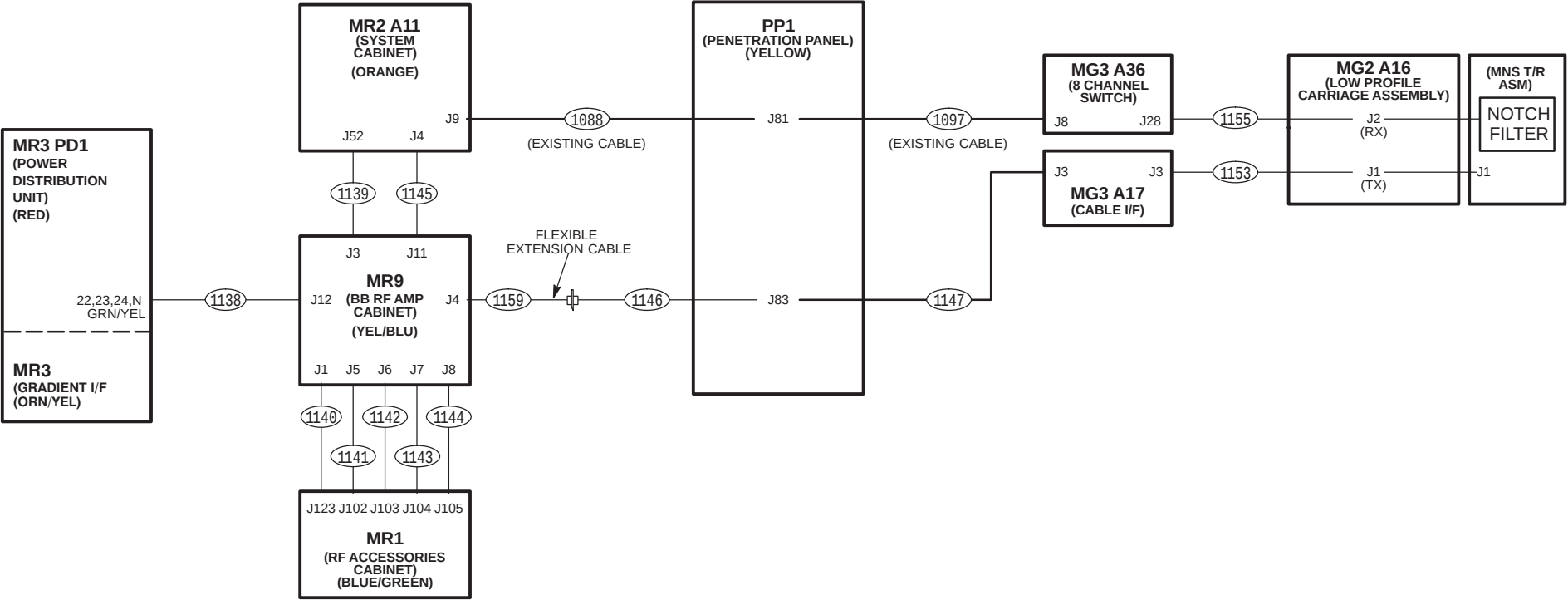
SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



SECTION 5 – PENETRATION PANEL

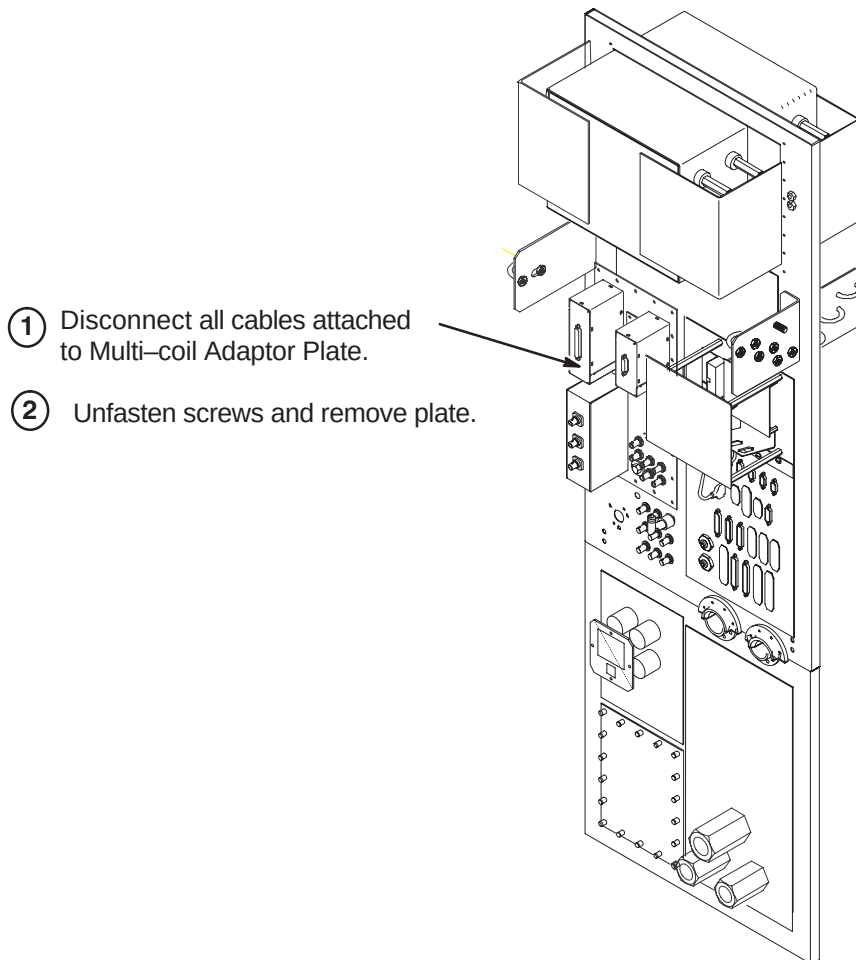
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

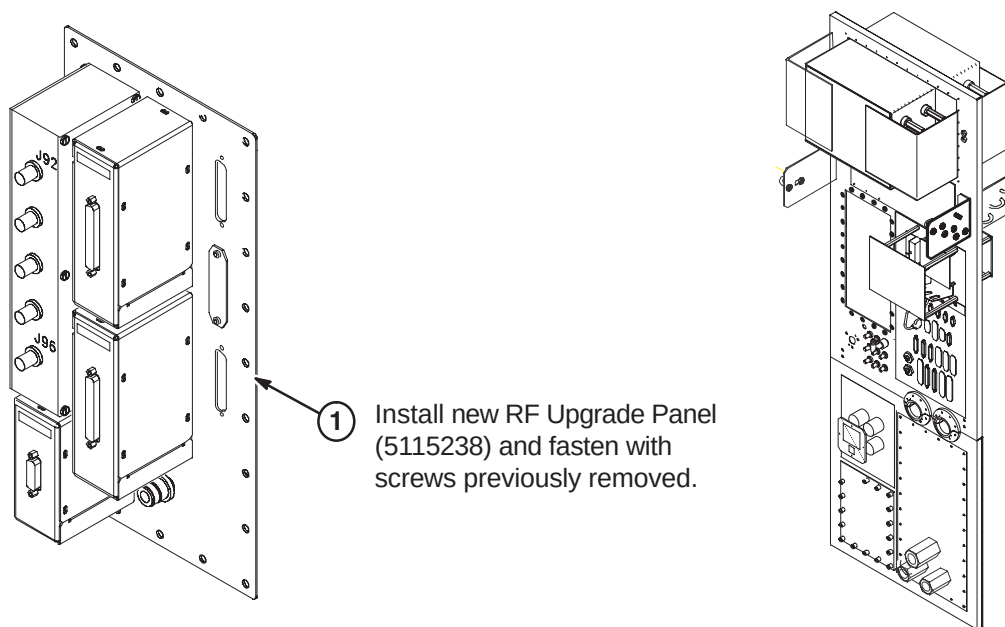
5-1 INTRODUCTION

The Multi-coil Adaptor Plate will be replaced by the RF Upgrade Panel (5115238).

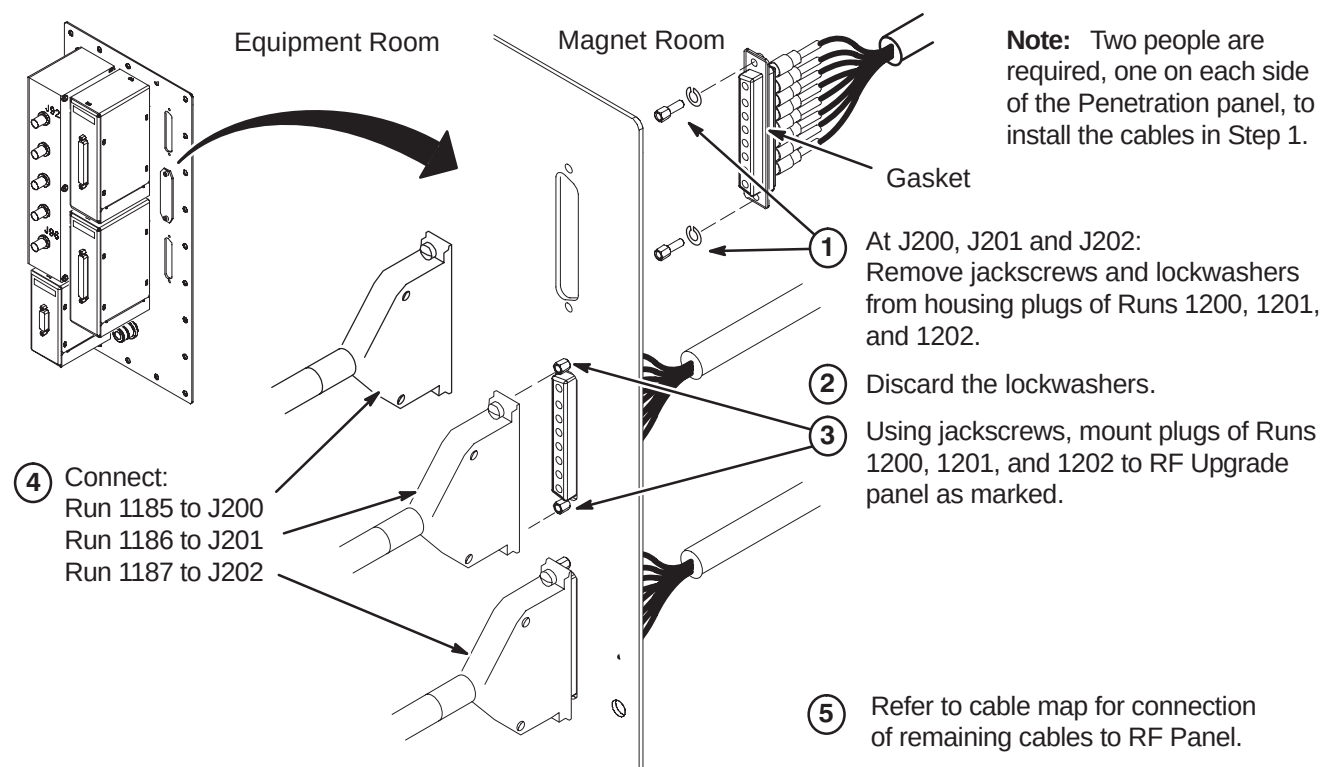
5-2 REMOVE MULTI-COIL ADAPTOR PLATE FROM PENETRATION PANEL



5-3 INSTALL NEW EXCITE RF PANEL



5-4 INSTALLATION OF CABLES AT J200, 201, AND J202



SECTION 6 – MAGNET ROOM

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6-2-4 DE-INSTALL PAC MODULE	6-6
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6-3-3 INSTALL 16 CHANNEL SWITCH	6-8
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6-3-6 INSTALL LPCA AND CABLE TRACKS	6-12
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6-4-2 ROUTE, TERMINATE, AND CONNECT RF CABLES	6-20

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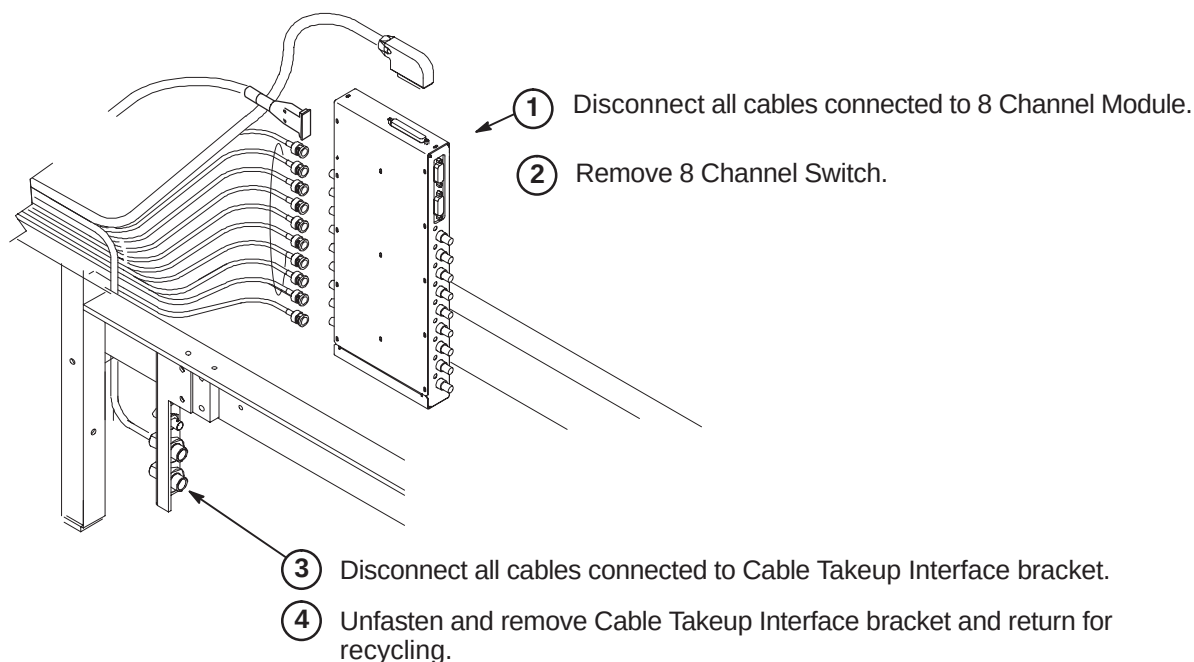
CAUTION

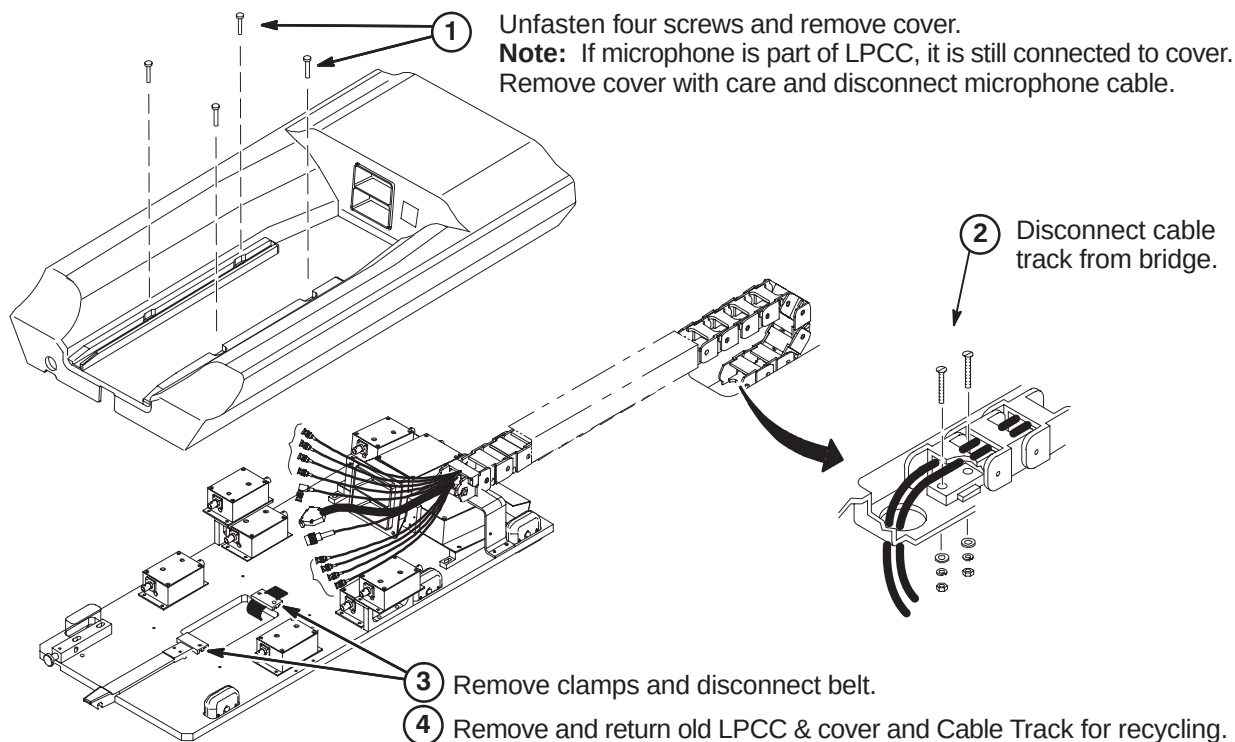
MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

6-1 INTRODUCTION

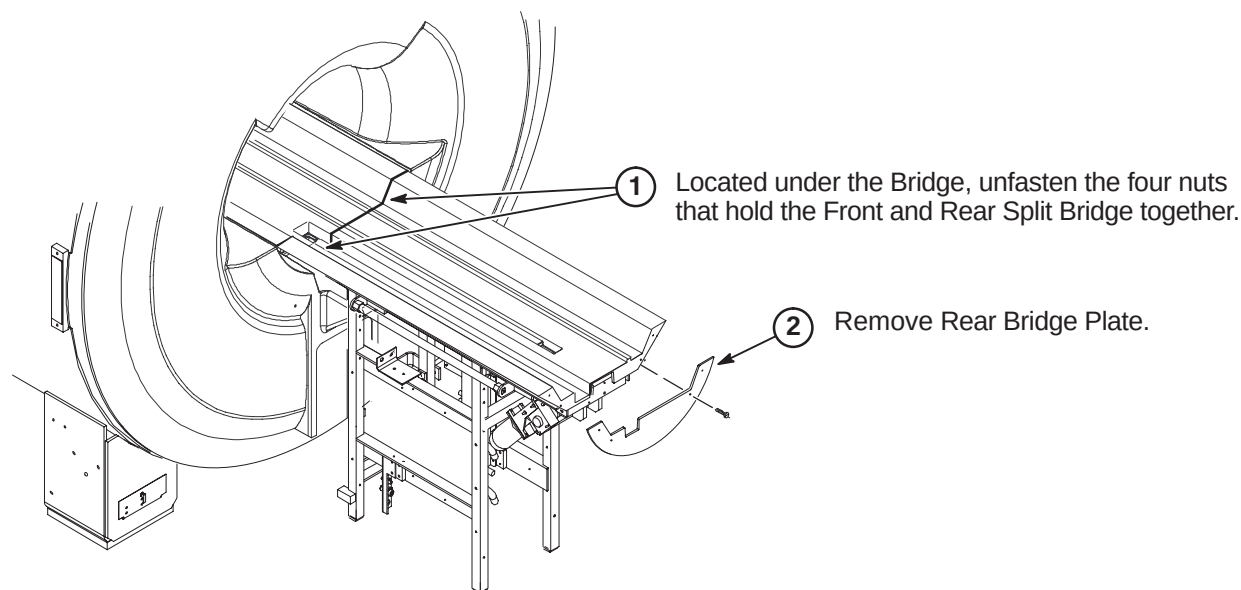
Modules on the Rear Pedestal, The Front and Rear Split Bridge, LPCA and Cable Track, SRI, and PAC will be replaced.

Several cables will be removed and replaced with new cables, some cables will be relabeled and reused, and new cables will be routed and connected. Refer to Section 4 cable maps for information on removing and installing cables

6-2 DE-INSTALLATION PROCEDURES**6-2-1 DISCONNECT CABLE TRACK CABLES FROM REAR PEDESTAL MODULES**

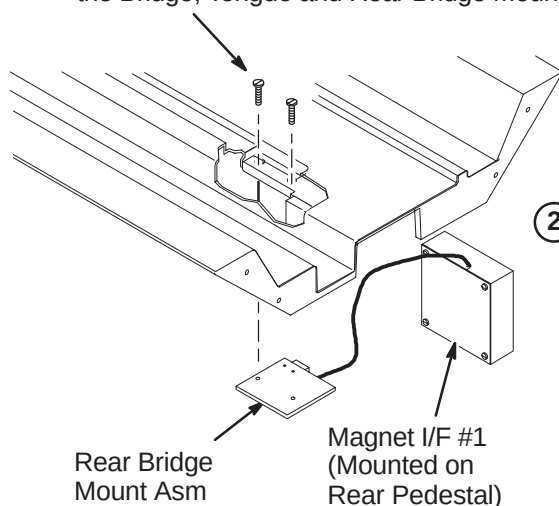
6-2-2 DE-INSTALL EXISTING LOW PROFILE CARRIAGE COVER & CABLE TRACK**6-2-3 DE-INSTALL BRIDGE**

Upgrading to EXCITE TwinSpeed Release 12.x requires a new Front and Rear Split Bridge.

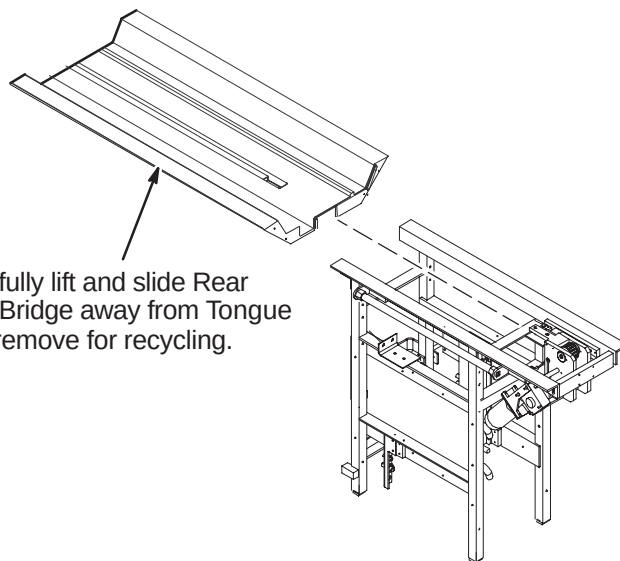
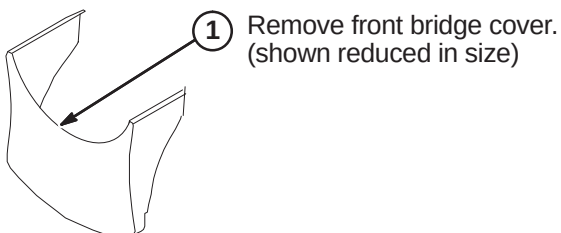
6-2-3-1 Unfasten Rear Split Bridge

6-2-3-2 Remove Rear Split Bridge

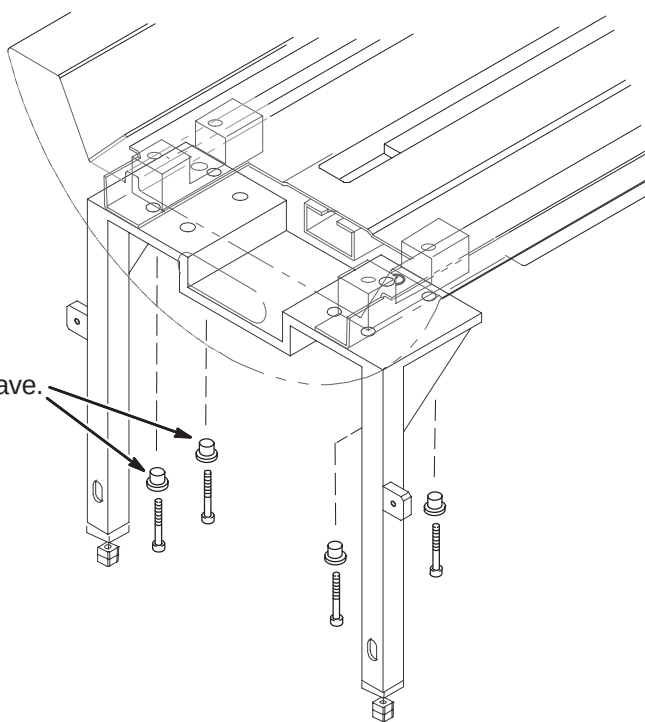
- ① Unfasten and remove the two screws connecting the Bridge, Tongue and Rear Bridge Mount Asm.



- ② Carefully lift and slide Rear Split Bridge away from Tongue and remove for recycling.

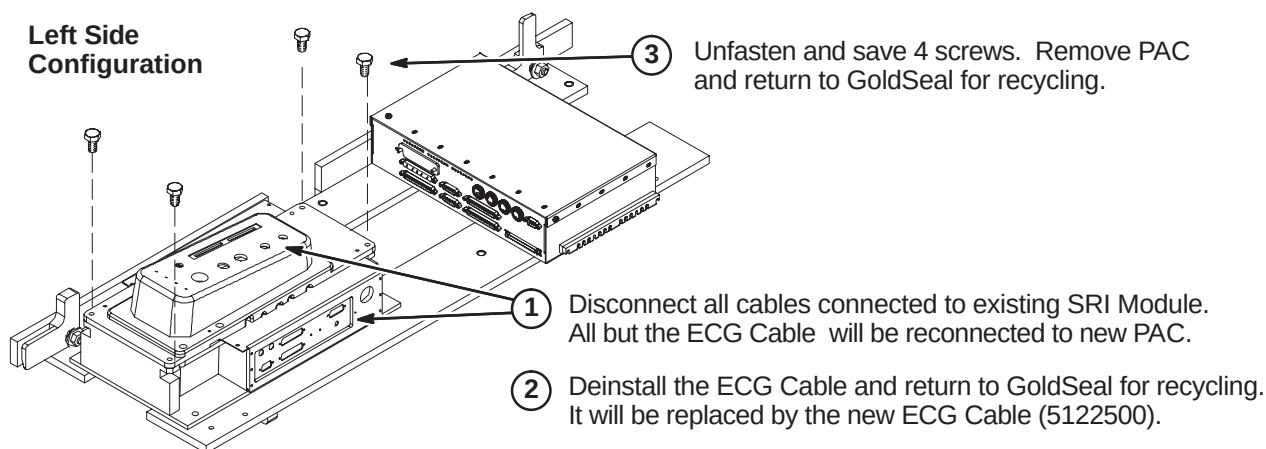
**6-2-3-3 Remove Front Split Bridge**

- ② Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings and save. These will be reused for installing the new Bridge.
- ③ Carefully lift and remove Front Split Bridge and return for recycling.



6-2-4 DE-INSTALL PAC MODULE

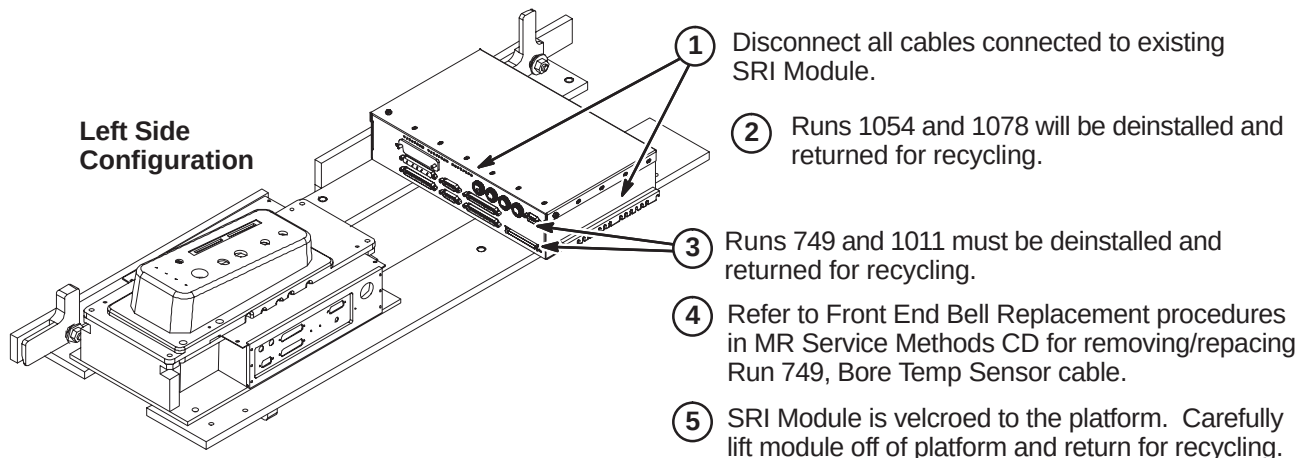
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6-2-5 DE-INSTALL SRI MODULE

The existing SRI Module will be replaced by the SRI-3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 749 (to be relabeled in the future as Run 1260) and 1011 (to be relabeled in the future as Run 1261). There is a functionality difference.

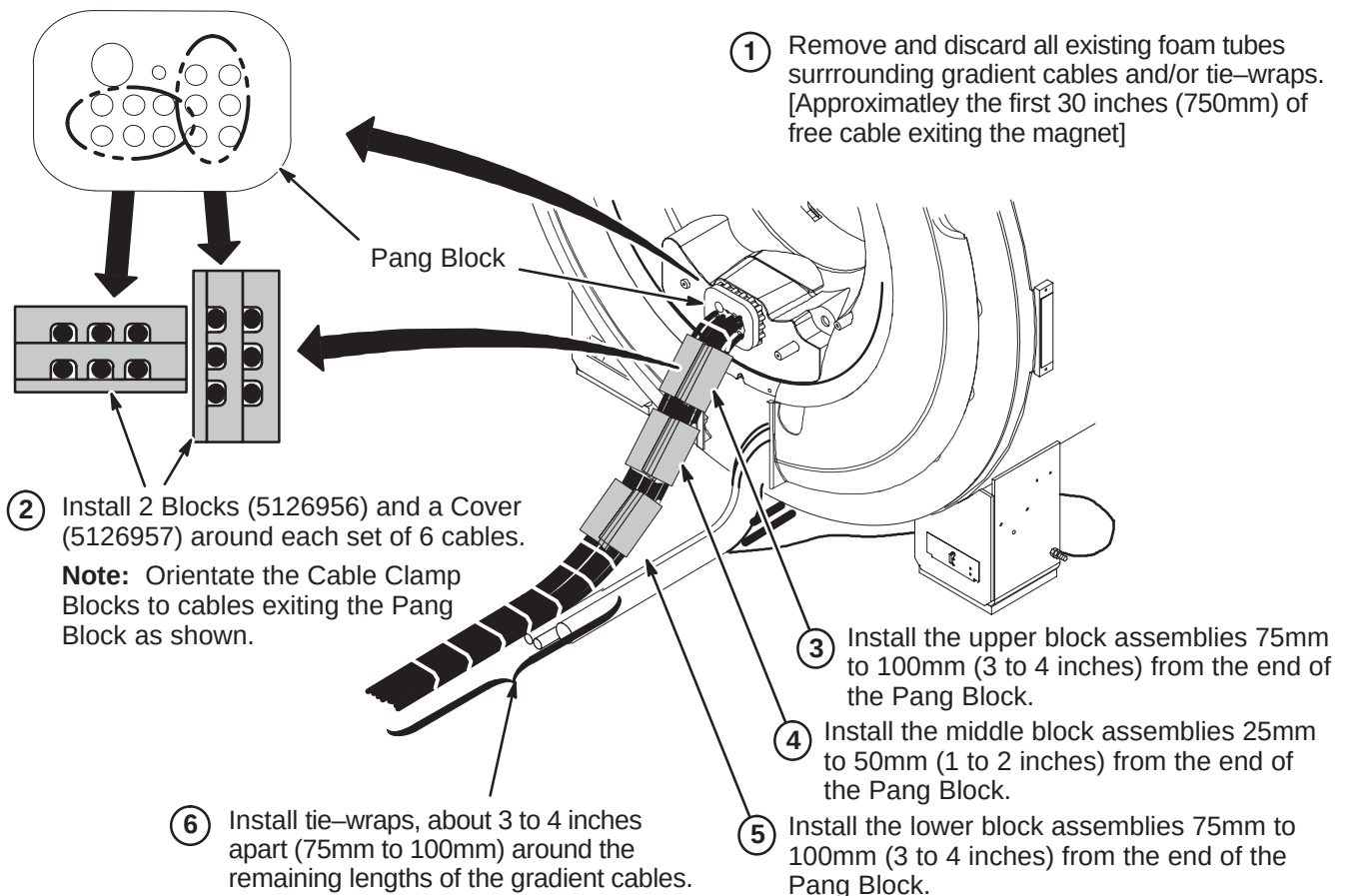


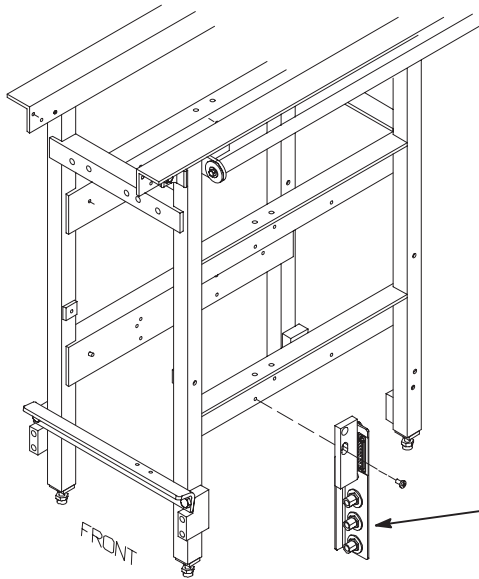
6-3 INSTALLATION OF NEW PARTS

6-3-1 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS

The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.

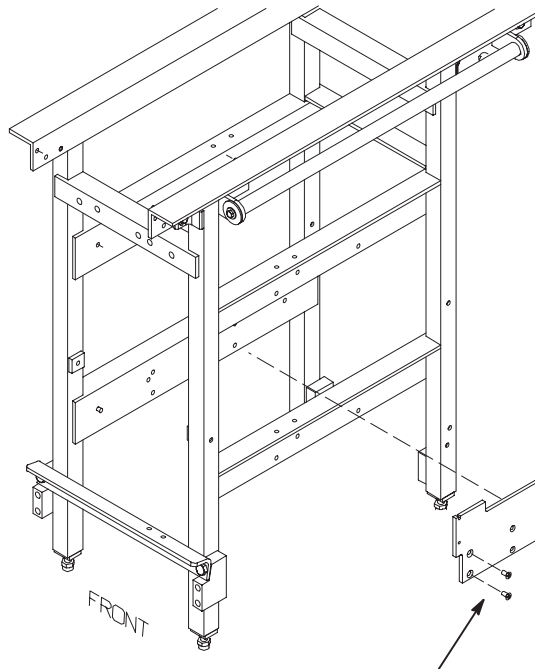
Note: Twelve blocks (5126956) and six covers (5126957) will be used, leaving six unused covers.



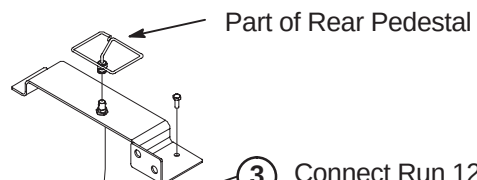
6-3-2 INSTALL NEW BULKHEAD INTERFACE

NOTE: Components and cables not shown on Rear Pedestal for clarity.

- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm pane Head Screws (2109873-24).

6-3-3 INSTALL 16 CHANNEL SWITCH

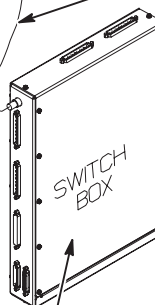
NOTE: Components and cables not shown on Rear Pedestal for clarity.



- ③ Connect Run 1204 (5116105) as shown.

- ① Install 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46-170686P2) and depending on style of Rear Pedestal:

- Four M6 x 12mm Slotted Flat Head Screws (25120369-86) in the four larger outside holes.
- or
- Two screws installed in the two center holes.

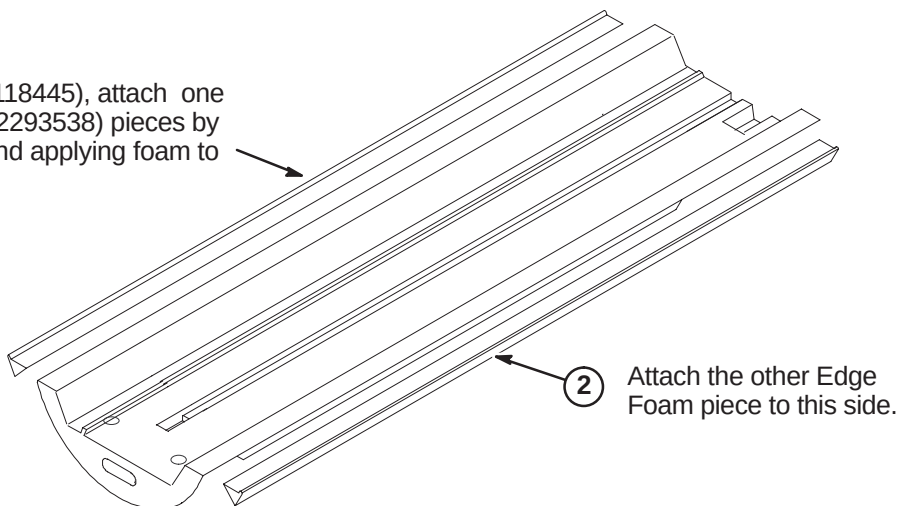


- ② Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46-170686P2) and four M3 x 12mm Pan Head screws (46-312342P24).

6-3-4 INSTALLATION OF FRONT BRIDGE

6-3-4-1 Attach Edge Foam

- ① Starting at the front of Bridge (5118445), attach one of the Twin Bridge Edge Foam (2293538) pieces by removing the protective cover and applying foam to bottom side of Bridge surface.

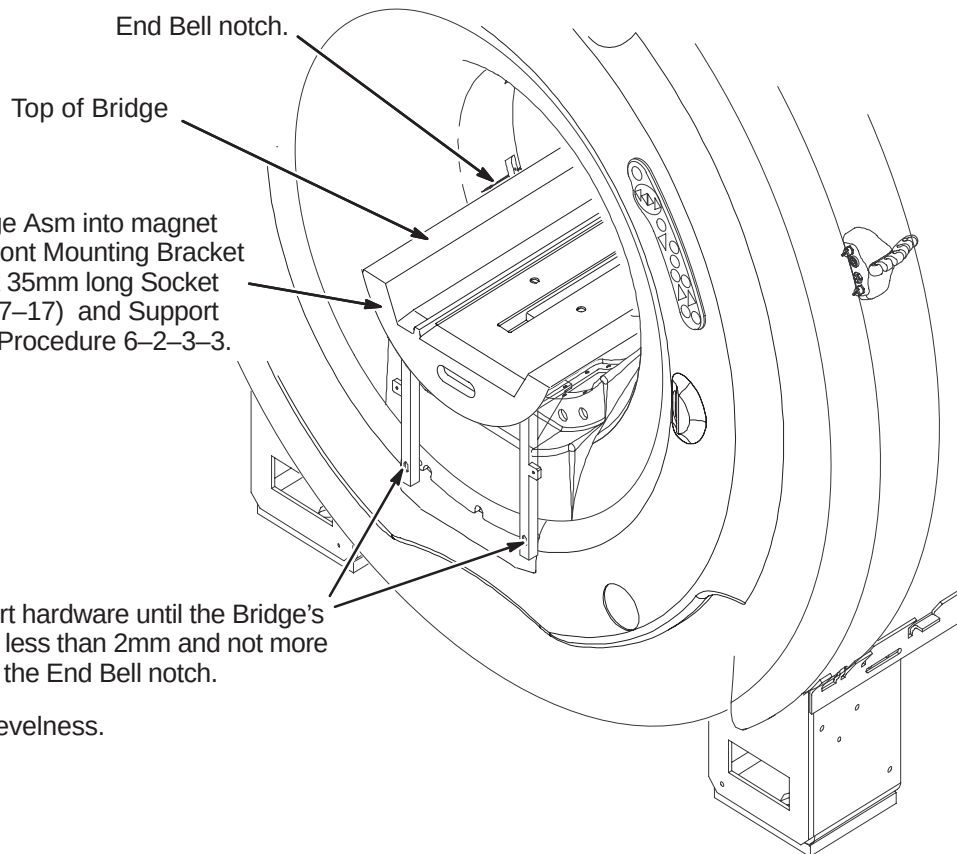


6-3-4-2 Install Bridge

- ① Install new Front Bridge Asm into magnet and attach Bridge to front Mounting Bracket using the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings removed in Procedure 6-2-3-3.

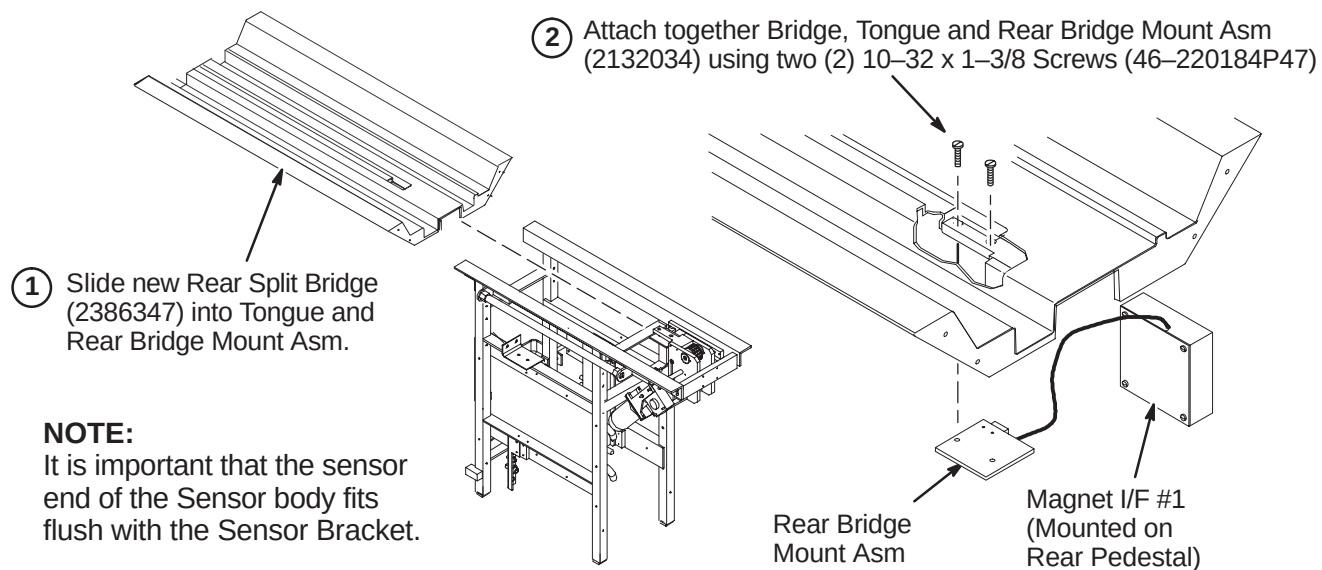
- ② Adjust the support hardware until the Bridge's top surface is not less than 2mm and not more than 4mm above the End Bell notch.

- ③ Check left-right levelness.

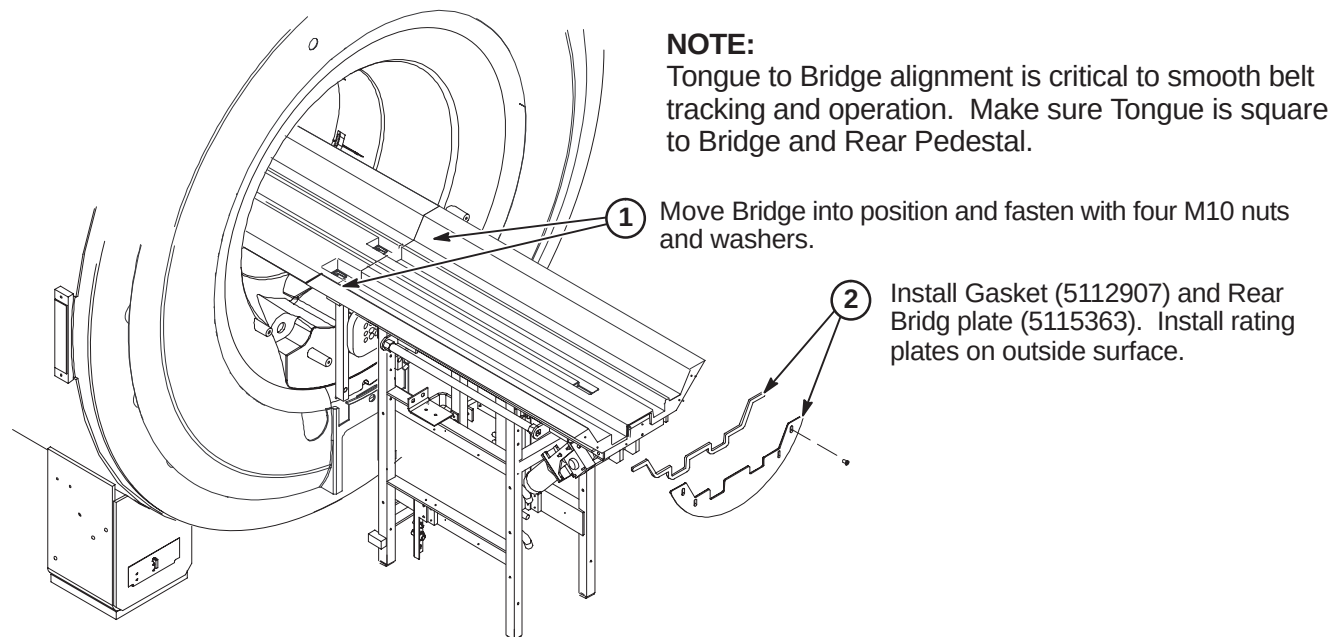


6-3-5 INSTALLATION OF REAR SPLIT BRIDGE

6-3-5-1 Install Rear Split Bridge on Rear Pedestal



6-3-5-2 Connect Rear Split Bridge to Front Split Bridge

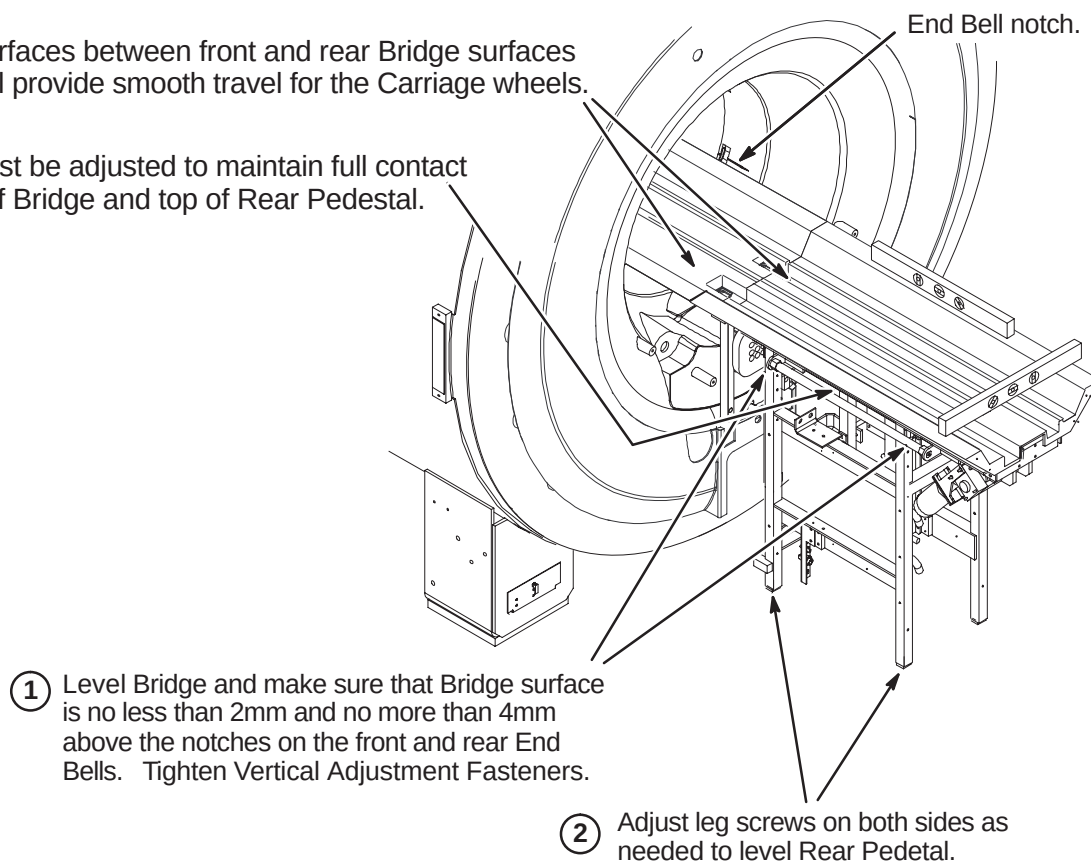


6-3-5-3 Level Bridge**NOTE:**

Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

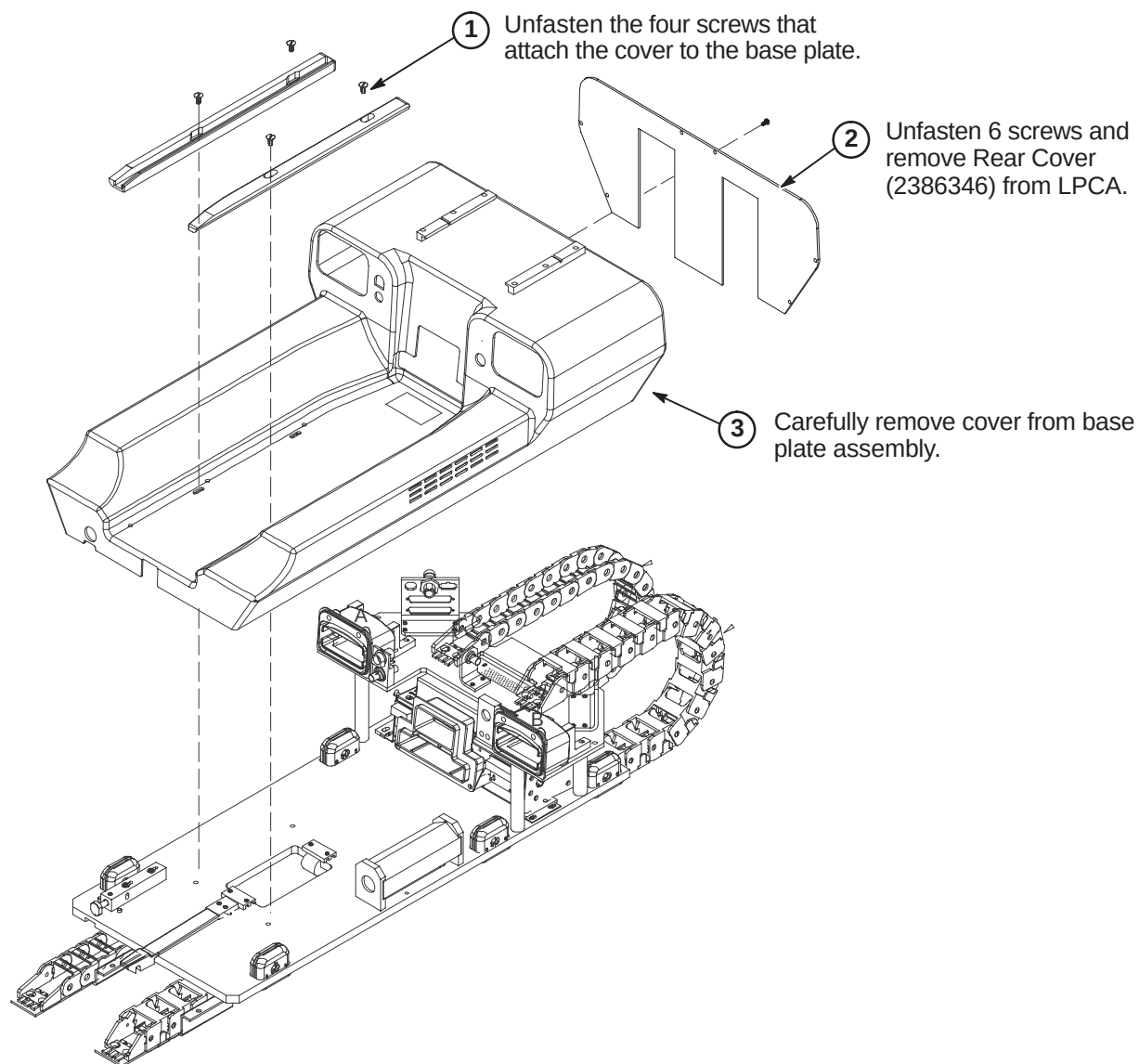
NOTE:

Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.



6-3-6 INSTALL LPCA AND CABLE TRACK(S)

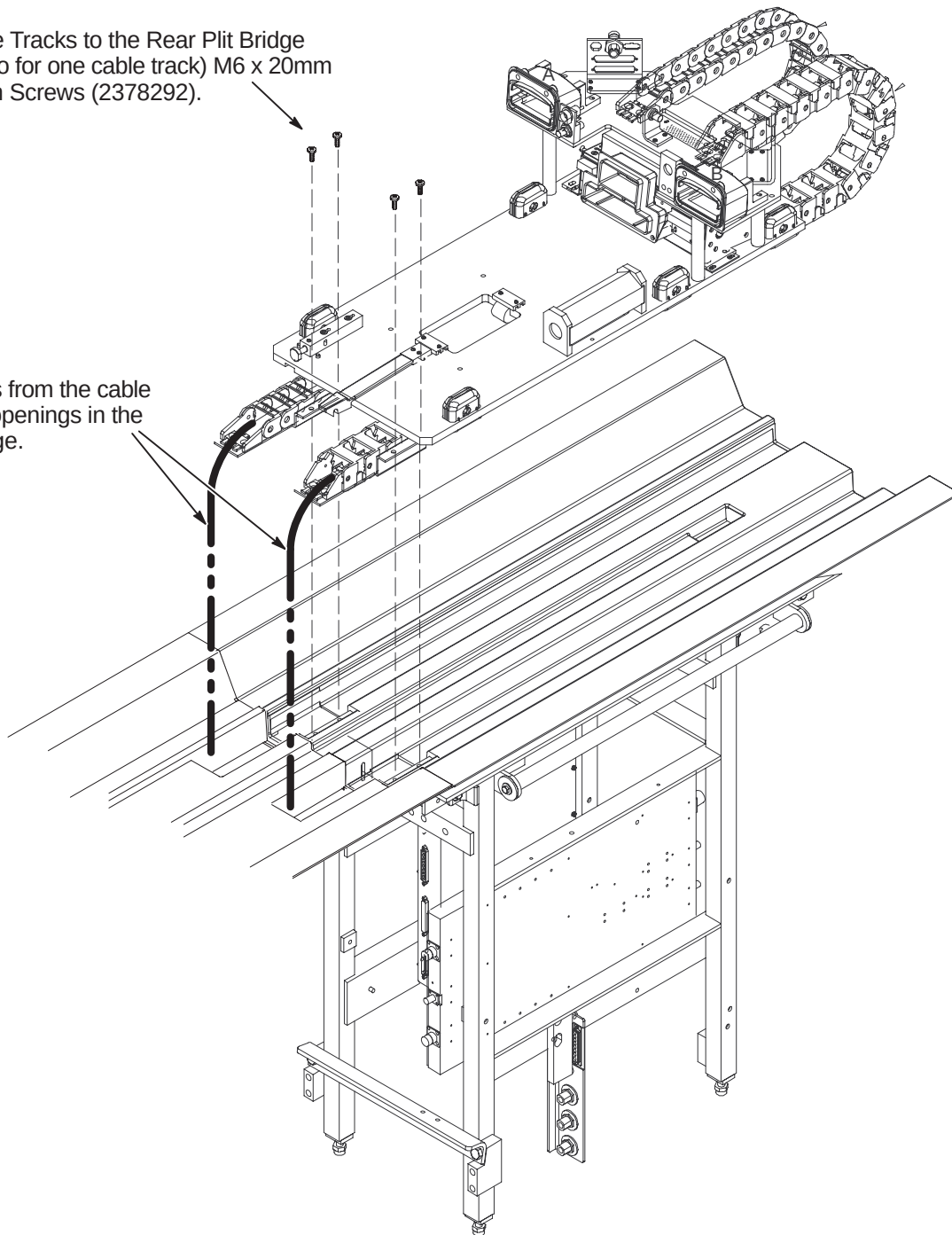
The WideOpen style enclosure will be upgraded to an LPCA with a dual cable track.

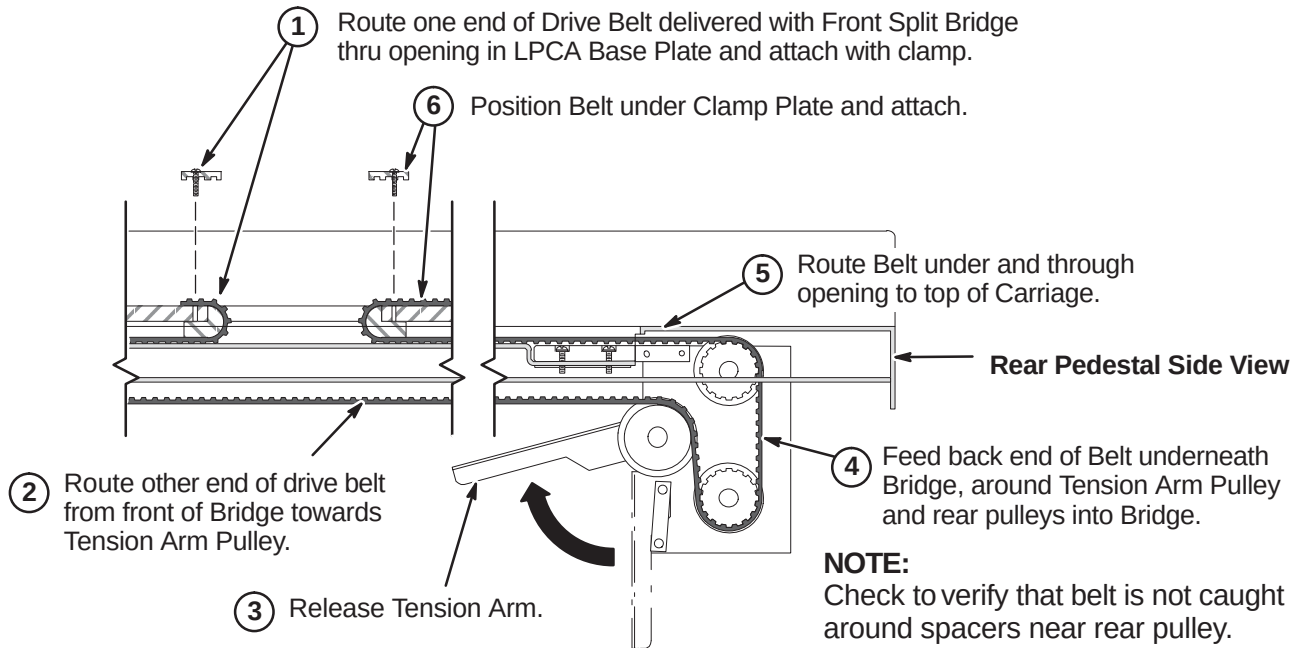
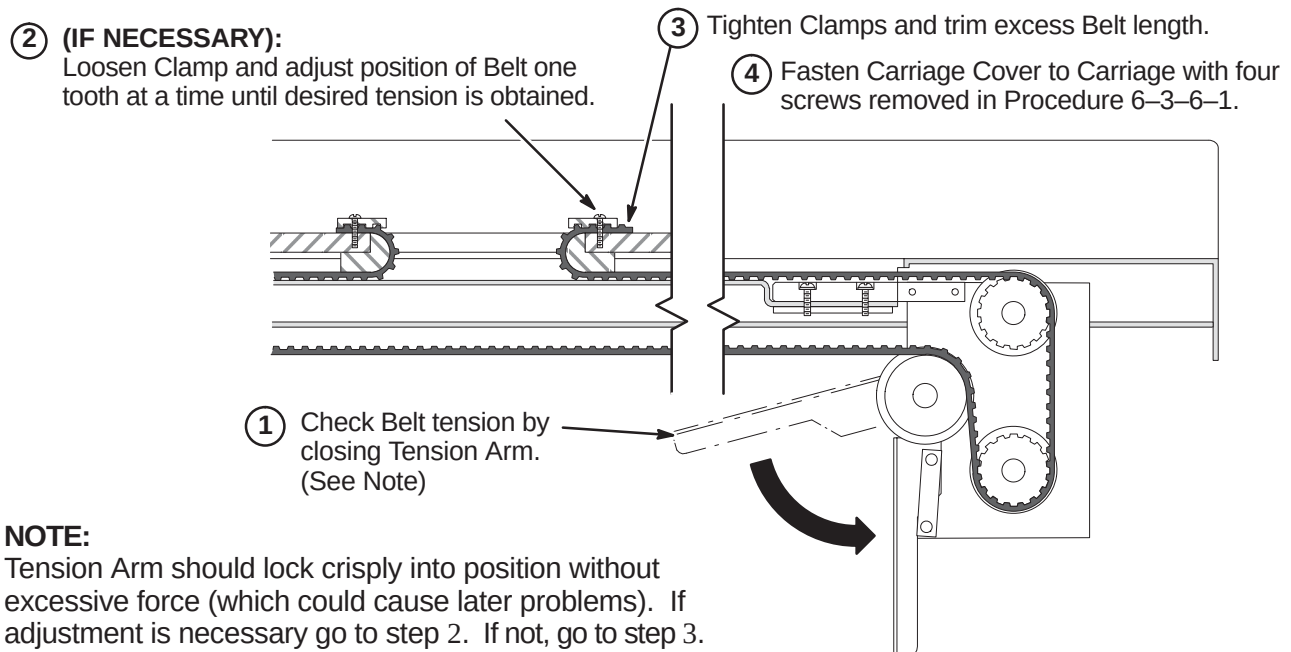
6-3-6-1 Remove LPCA Cover

6-3-6-2 Attach Cable Track(s) to Bridge

- ② Attach the Cable Tracks to the Rear Plit Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable tracks thru the openings in the Front Split Bridge.



6-3-6-3 Route and Attach Drive Belt**6-3-6-4 Final Adjustment of Drive Belt**

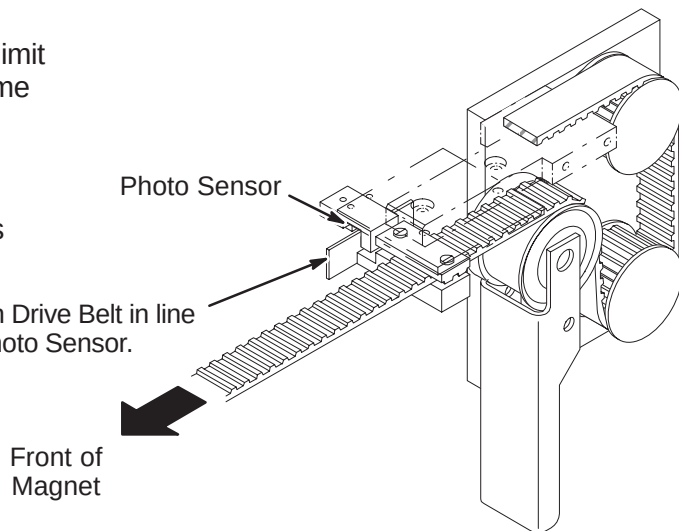
6-3-6-5 Installation of Sensor Flag**NOTE:**

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

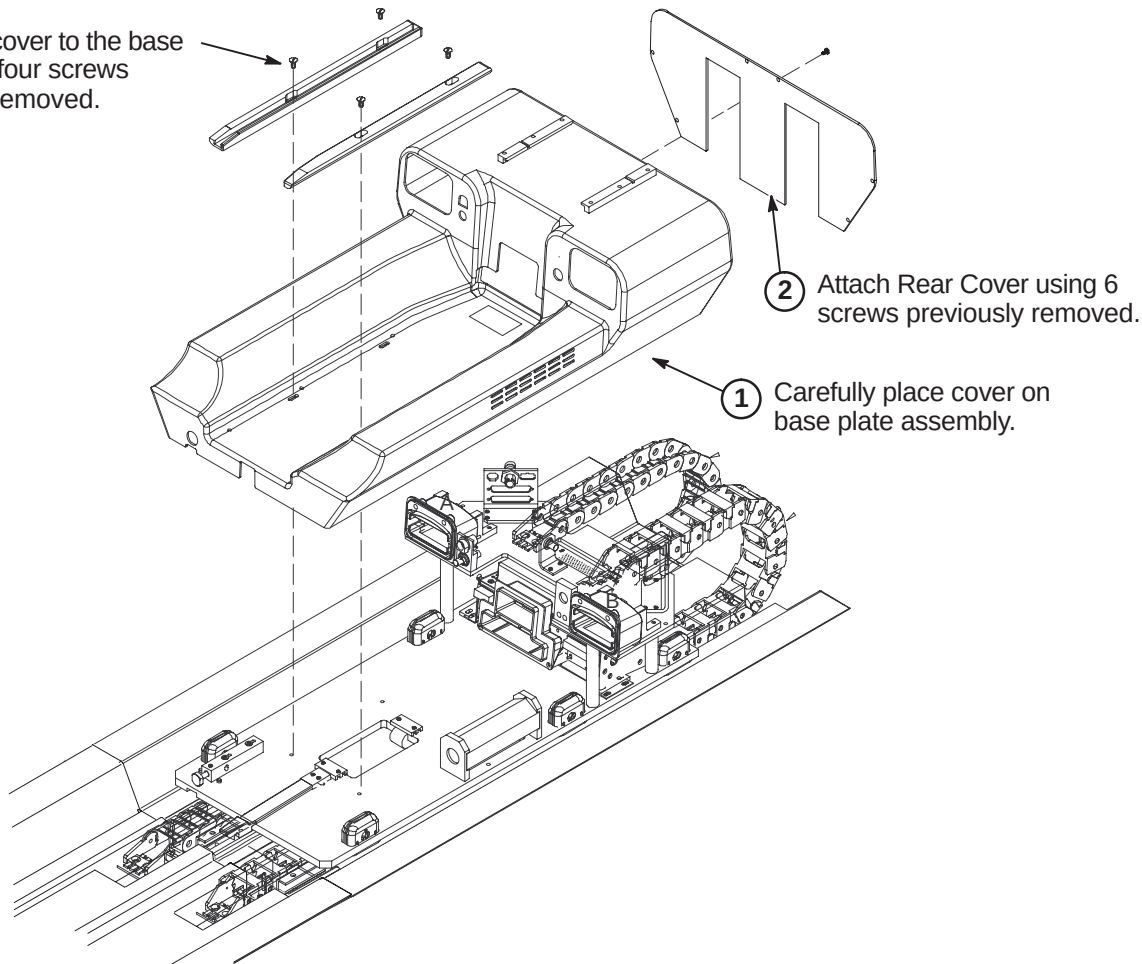
NOTE:

The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

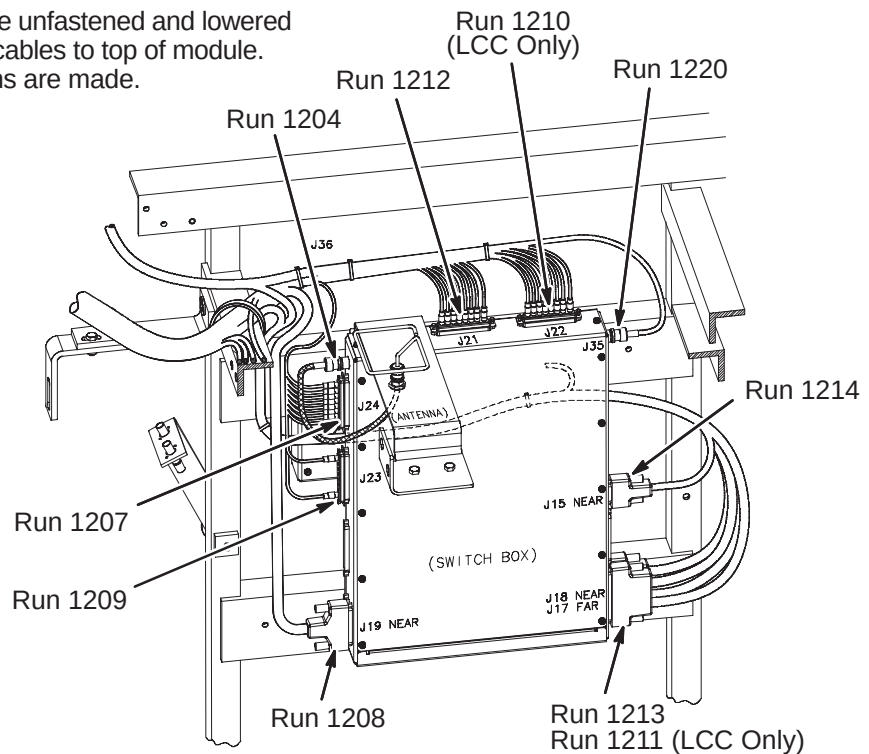
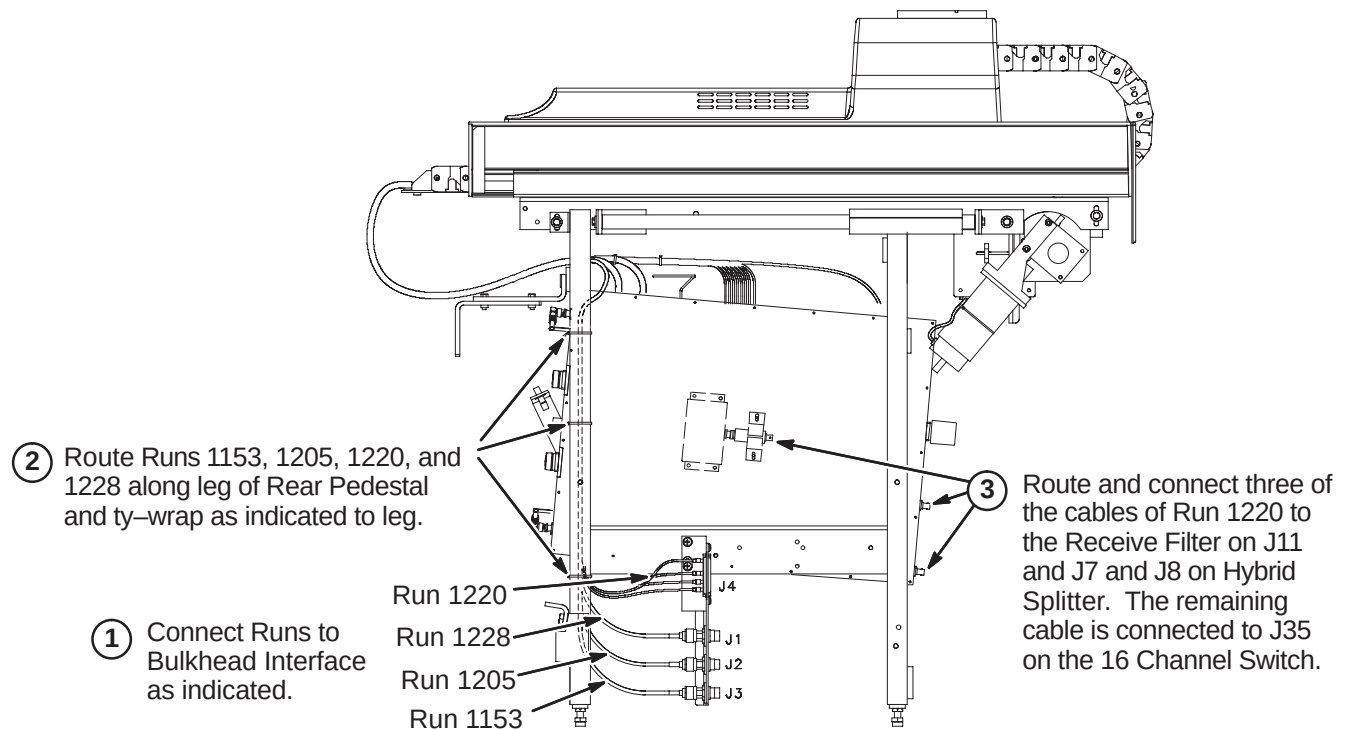
**6-3-6-6 Installation of LPCA Cover**

- ③ Attach the cover to the base plate using four screws previously removed.



6-3-6-7 Connect Cables from Cable Track to 16 Channel Switch Module

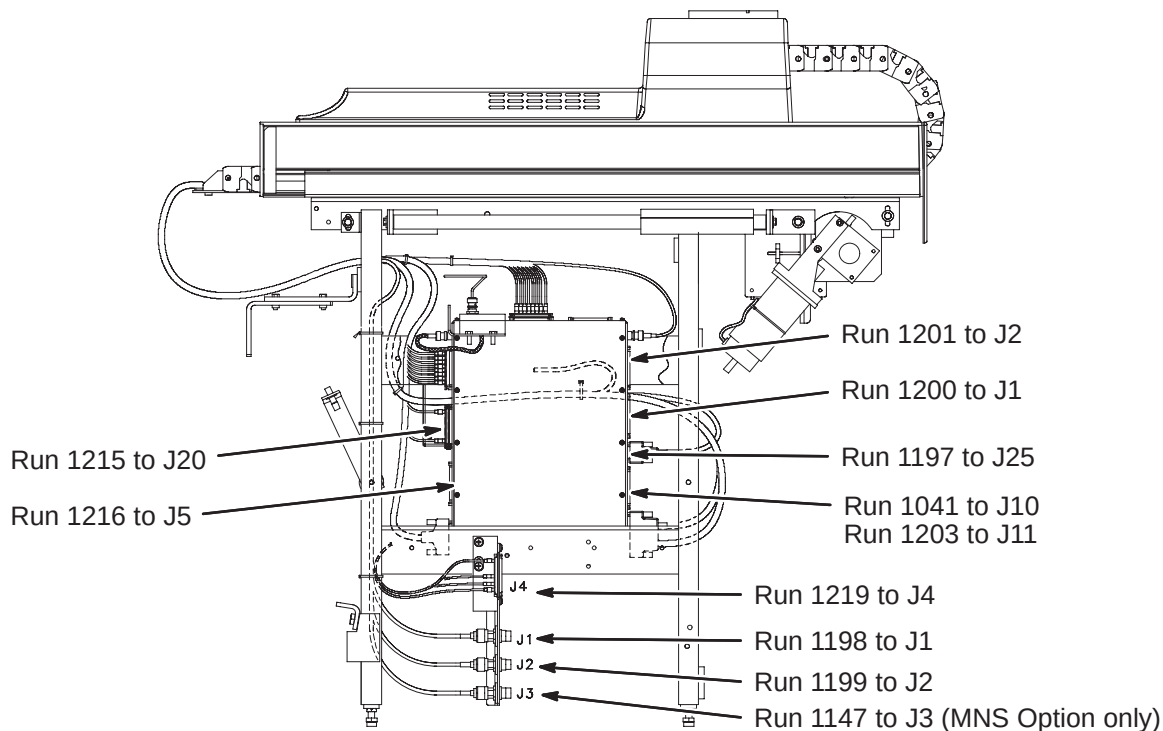
- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**6-3-6-8 Connect Cables from Cable Track to Bulkhead Interface**

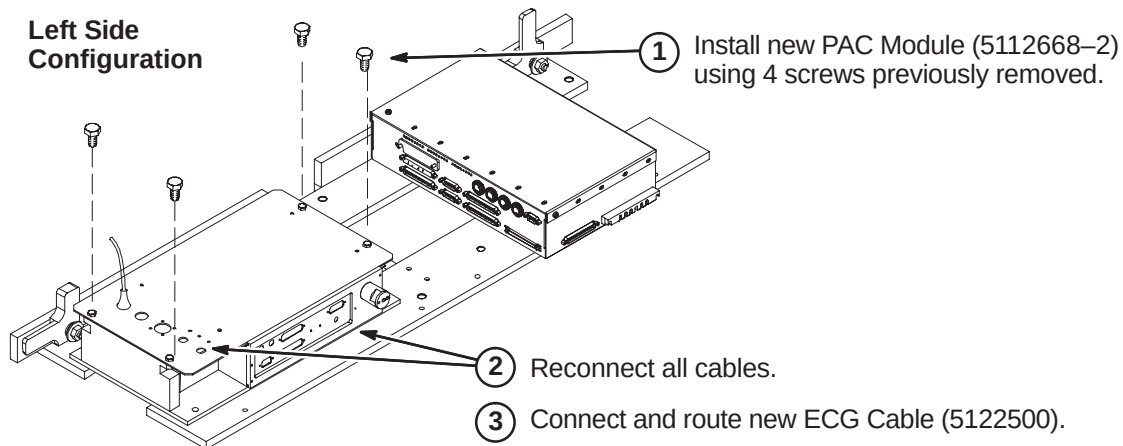
6-3-7 CONNECT REMAINING CABLES TO REAR PEDESTAL

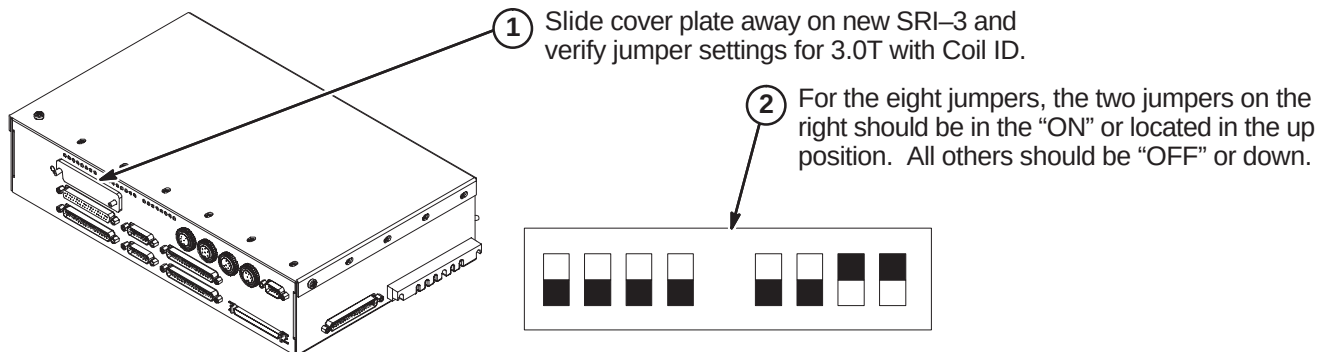
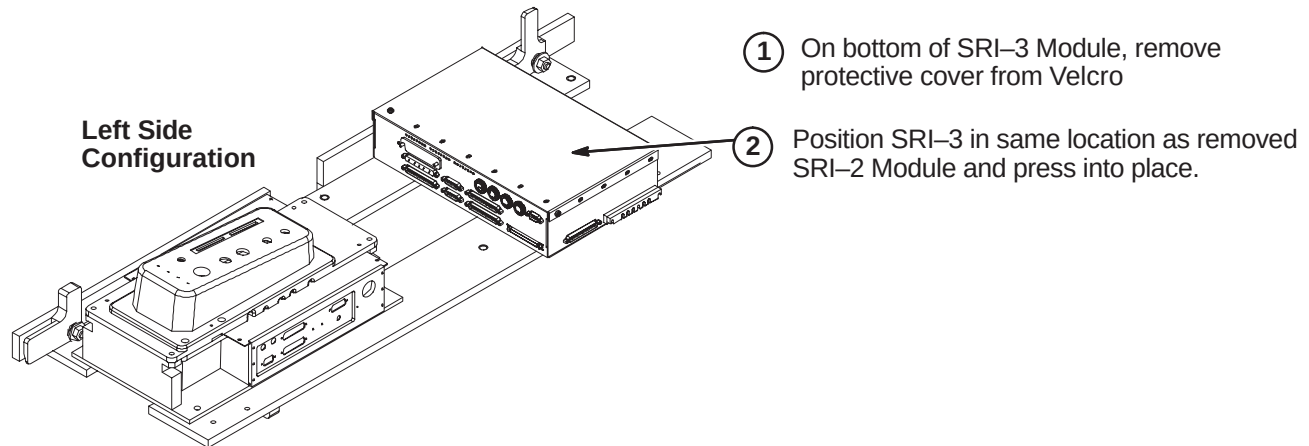
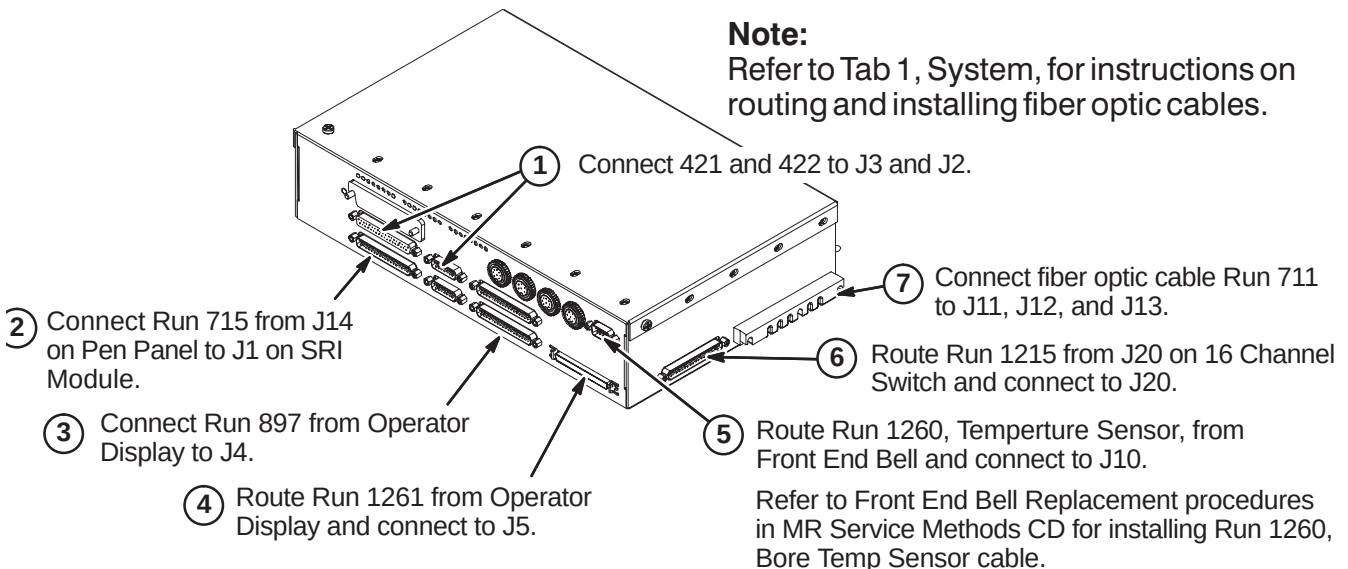
- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

Note: Refer to Section 5-4 for instructions on connecting Runs 1200, 1201, and 1202 to Penetration Panel.

**6-3-8 INSTALL NEW PAC MODULE AND ECG CABLE**

Reverse the deinstallation steps for installation of the new PAC Module. Also install the new ECG Cable.



6-3-9 INSTALL SRI-3 MODULE**6-3-9-1 Verify SRI-3 Jumper Settings****6-3-9-2 Position SRI-3 Module****6-3-9-3 Connect Cables to SRI-3 Module**

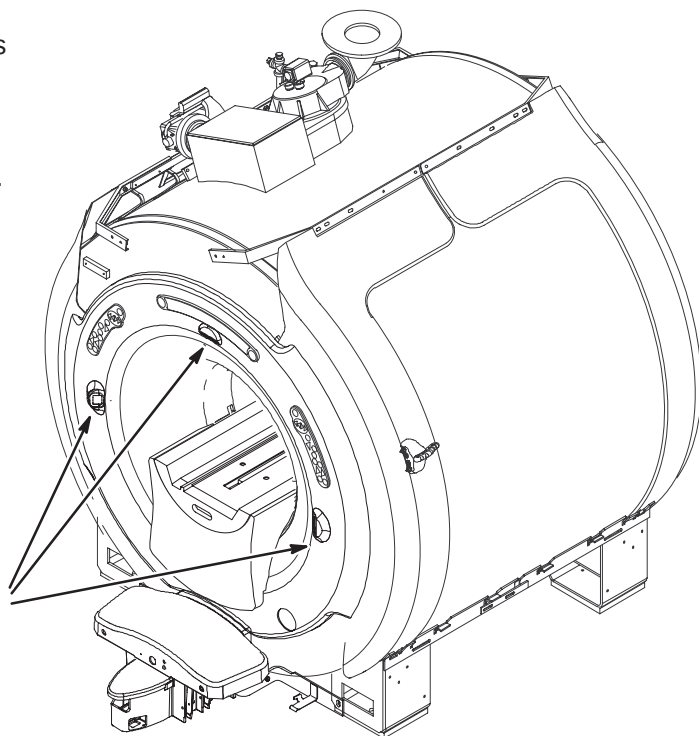
6-3-10 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



English



- ① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



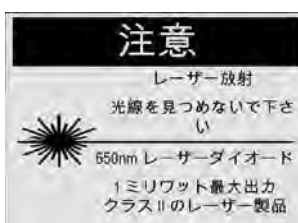
Spanish



Italian



Swedish

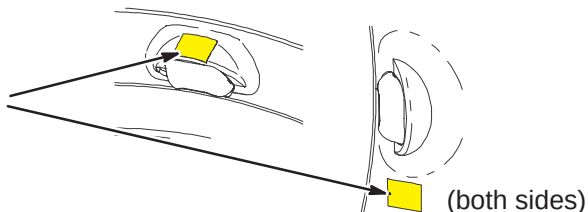


Japanese



Chinese

- ② If enclosure does not have labels attached, attach new labels as indicated.



6-4 TIMES MICROWAVE RF CABLES INSTALLATION

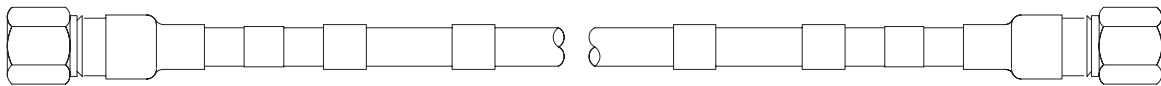
6-4-1 CABLE INFORMATION

Two cables are to be created and other associated parts to be installed are provided for this procedure. Also provided is a Times Microwave stripping tool for the LMR 600 (2352193), and a Times Microwave cutting tool (5111565).



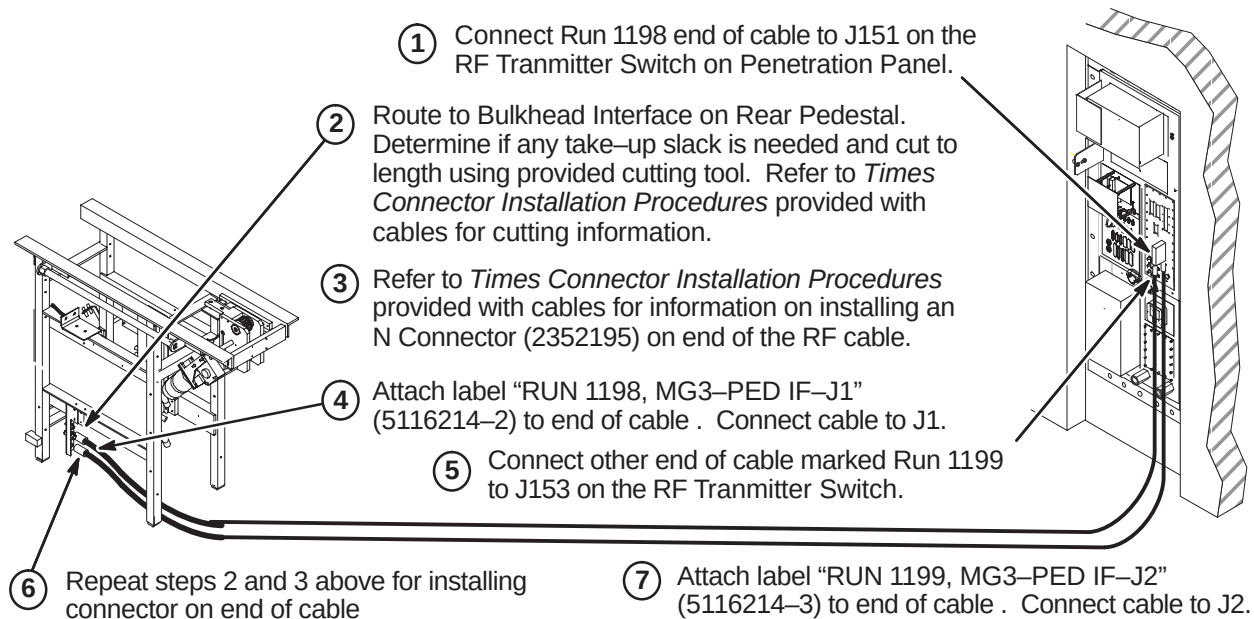
Do not subject LMR-600FR cable to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Cable 5115915, will be cut to length to create two cables, Runs 1198 and 1199, and terminated for routing in the scan room. The remaining connectors, fasteners, and labels will be installed as directed in the following steps.



Run 1198 and Run 1199

6-4-2 ROUTE, TERMINATE, AND CONNECT RF CABLES



SECTION 7 – OPERATOR WORKSPACE

7-1 INTRODUCTION

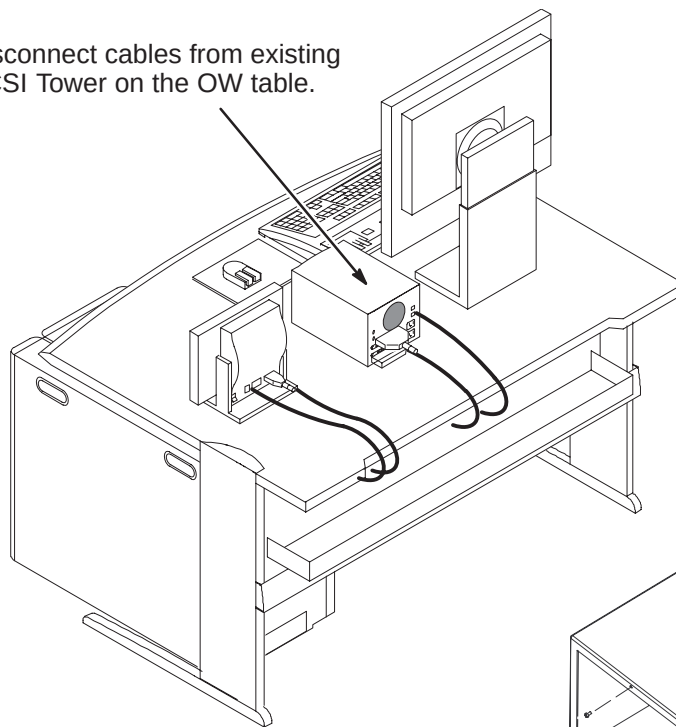
A new DVD–RW will be installed in the existing SCSI Tower on the Operator Workspace Table.

CAUTION

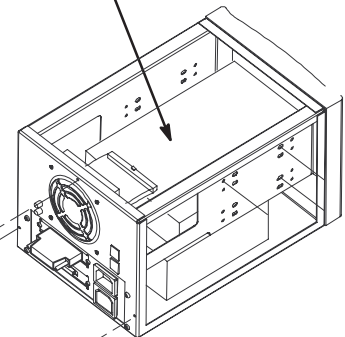
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

7-2 PREPARE SCSI TOWER FOR UPGRADE

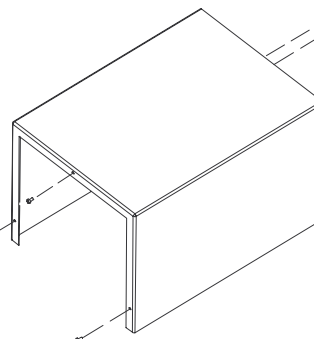
- ① Disconnect cables from existing SCSI Tower on the OW table.

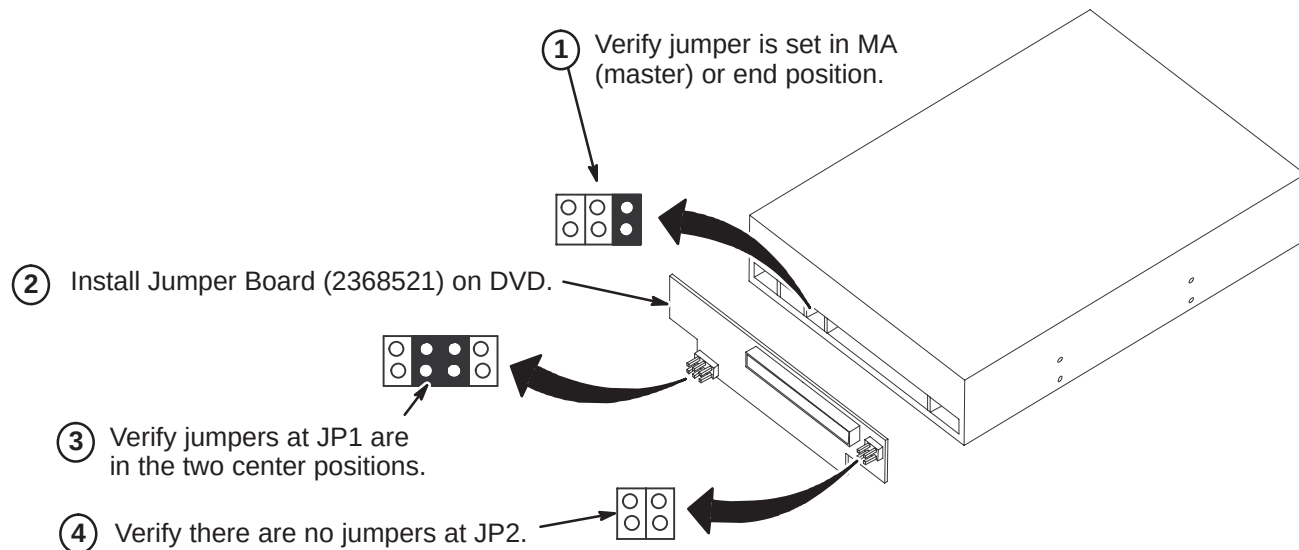
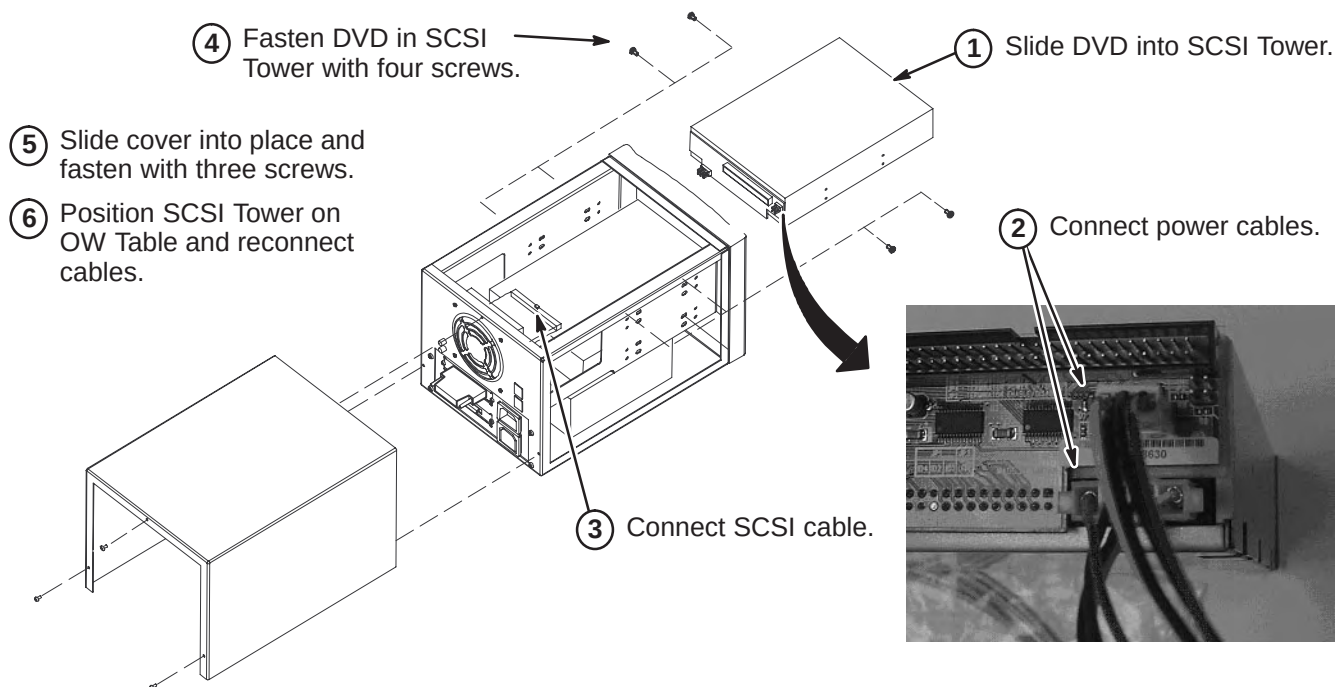


- ③ If installed in top position, move MOD Drive to lower position. Reconnect to ribbon cable.



- ② Unfasten three screws on back of SCSI Tower cover and slide off cover.



7-3 VERIFY JUMPER POSITIONS ON DVD**7-4 INSTALL DVD INTO SCSI TOWER**

SECTION 8 – SYSTEM CABINET

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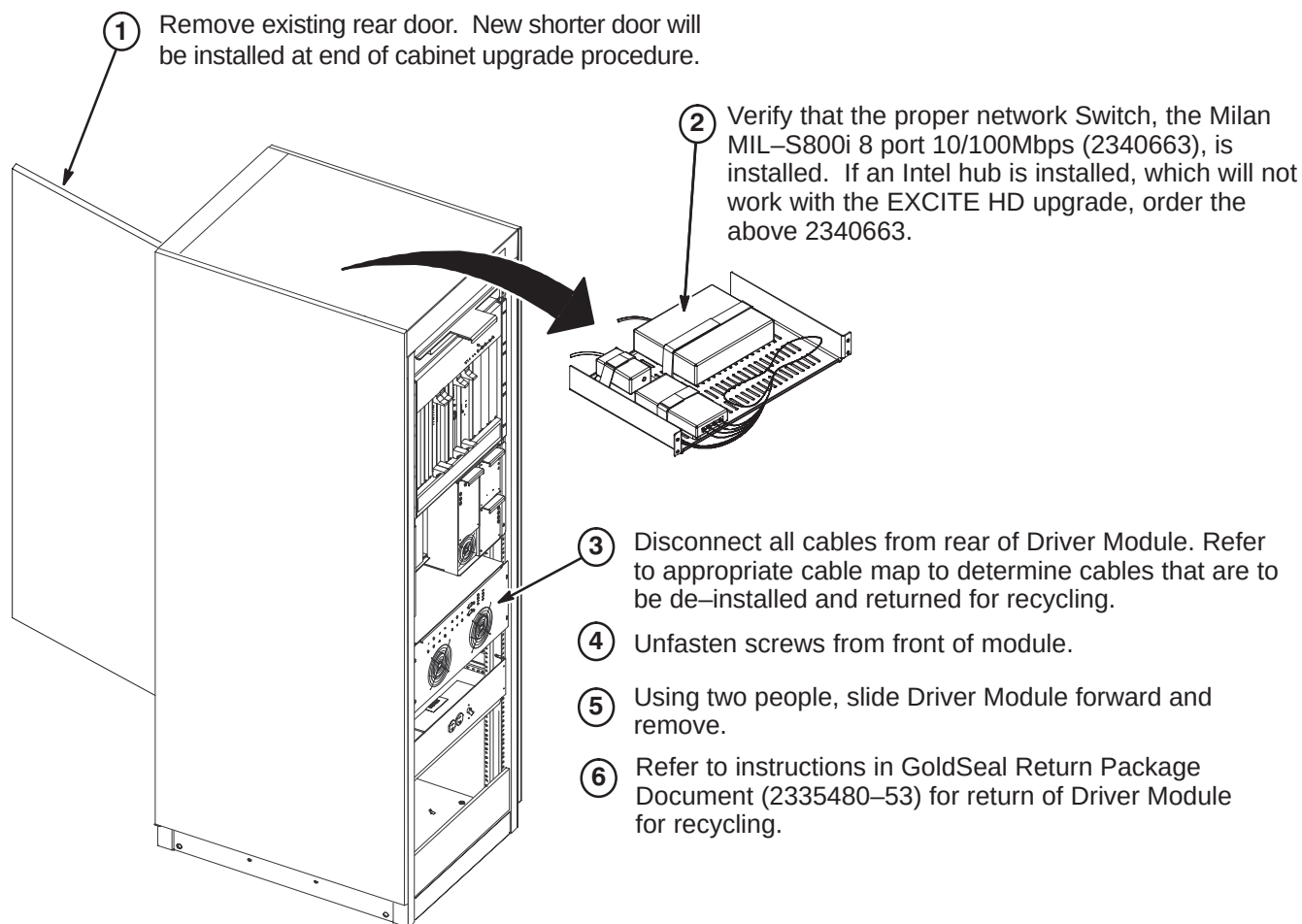
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CAUTION

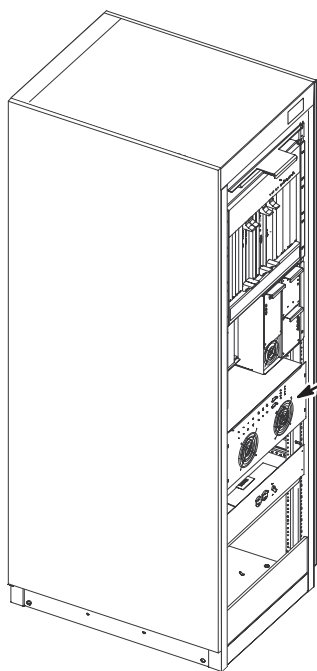
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

8-1 INTRODUCTION

The System cabinet will be modified for the upgrade to EXCITE Release 12.x. Refer to cable maps in Section 4 for disconnection and removal of existing cables and installation of new cables. Refer to the following steps for the upgrade procedures that pertain to your site.

8-2 PREPARE SYSTEM CABINET FOR UPGRADE

8-3 INSTALL NEW DRIVER MODULE



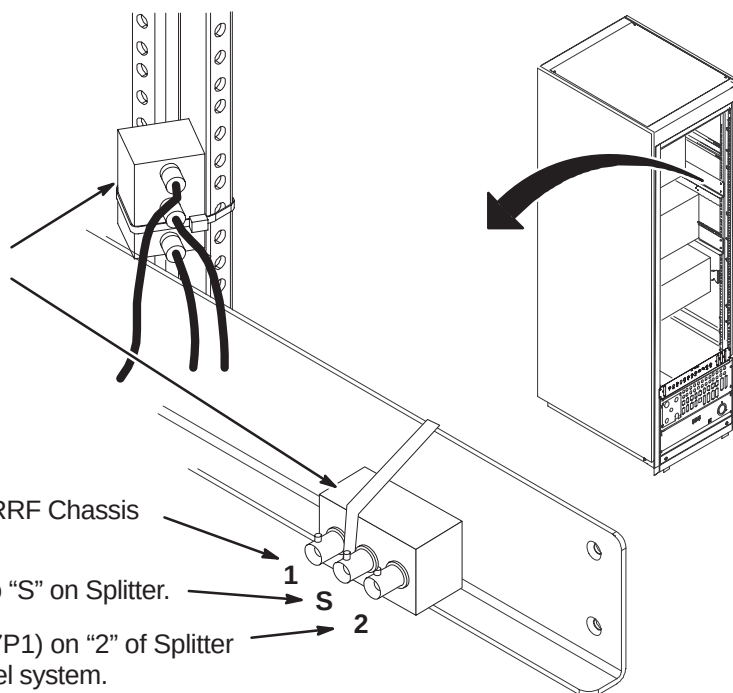
- ① Using two people, slide new Driver Module into position.
- ② Secure in place with previously removed screws.
- ③ Refer to applicable cable map for connection of cables.

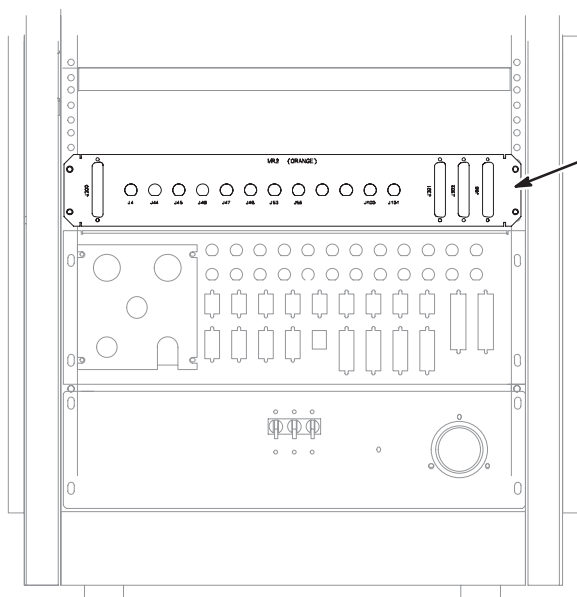
8-4 INSTALL COAX POWER SPLITTER

- ① Install BNC Coax Power Splitter (5115474) on vertical rail **OR** upper "L" bracket. Fasten with ty-wrap.

- ② Connect cable from slot 12R J31 at rear of RRF Chassis to "1" on Splitter.
- ③ Connect cable from Interface Panel J202 to "S" on Splitter.
- ④ Install 50 Ohm BNC Terminator (46-296817P1) on "2" of Splitter if site is NOT being upgraded to a 16 channel system.

If 16 Channel is installed, connect cable from slot 3R J31 at rear of RRF Chassis to "2" on Splitter.



8-5 INSTALL NEW INTERFACE PANEL

① At rear of cabinet, install 1.5T upgrade Interface Panel (5119238) above existing interface panel and fasten with four (4) M5 x 16mm Pan head screw (2279742-2).

② Use table below for connecting the remaining cables in the System Cabinet. All the cables connected to the new I/O Panel are pre-installed on the I/O Panel.

CABLE PART #	ONE END	OTHER END
5117295	Driver Module J4	MGD Slot 11R J21
5117963	Driver Module J10	I/O Panel J66
2394870-3	RF Slot 14R J5	I/O Panel J200
2394870-4	RF Slot 5R J5	I/O Panel J201
5115416	Splitter S RF Slot 14R J13 I/O Panel J100 I/O Panel J101	I/O Panel J202

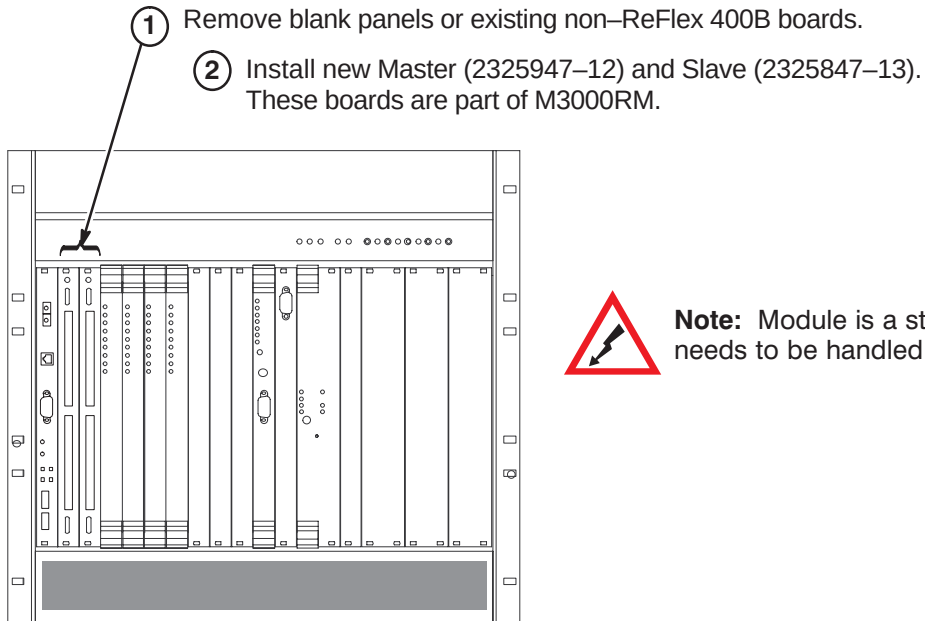
8-6 PROCEDURES FOR UPGRADING TO 8 CHANNEL

After upgrade, all systems will have one DRF2 board and IRF2 board.

8-6-1 INSTALL REFLEX 400 AP BOARDS (or Optional 800 AP Boards)

Upgrade to an 8 Channel system requires the installation of ReFlex 400 AP boards, M3000RM option. Refer to Direction 5121980, also part of M3000RM, for instructions on installing AP boards.

Note: If site already has ReFlex 400 AP boards installed, then do not perform this procedure.

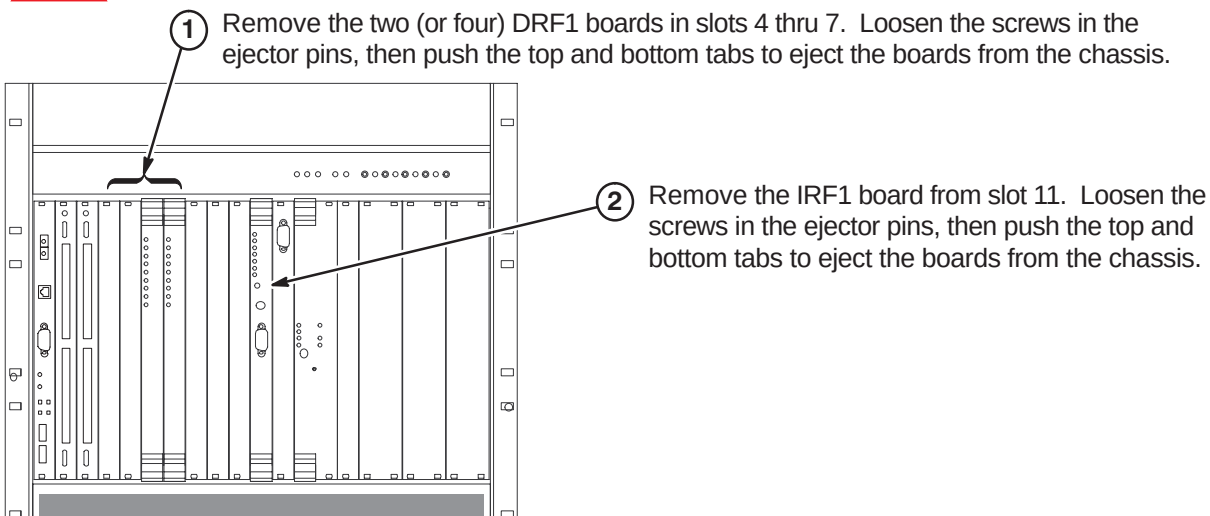


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

8-6-2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS

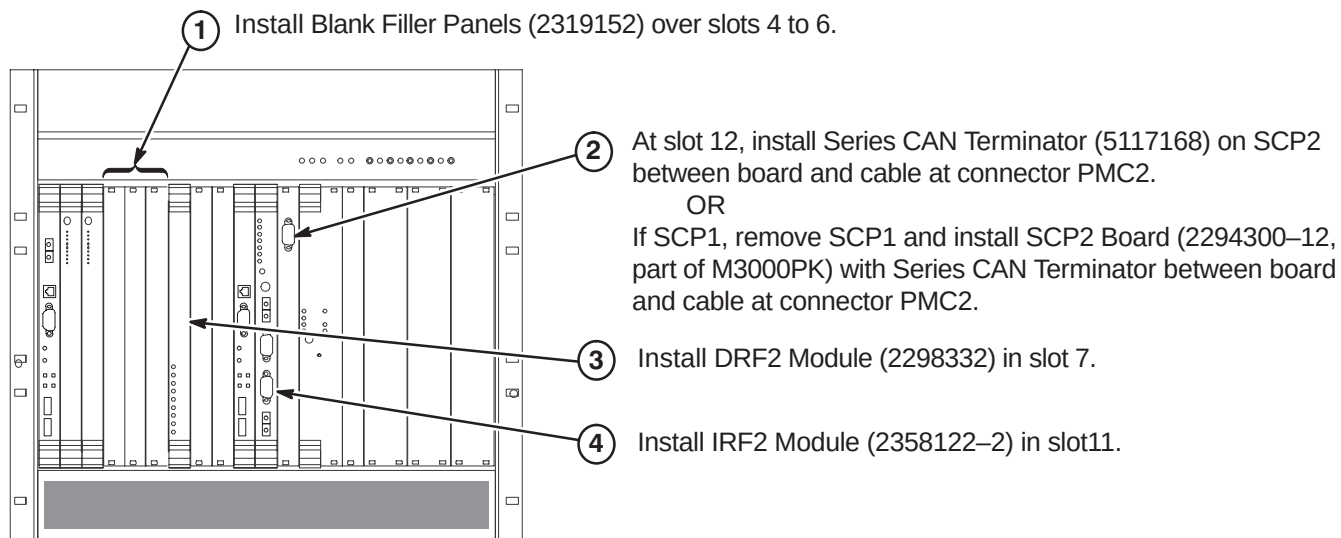


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



8-6-3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



8-7 PROCEDURES FOR UPGRADING TO 16 CHANNEL

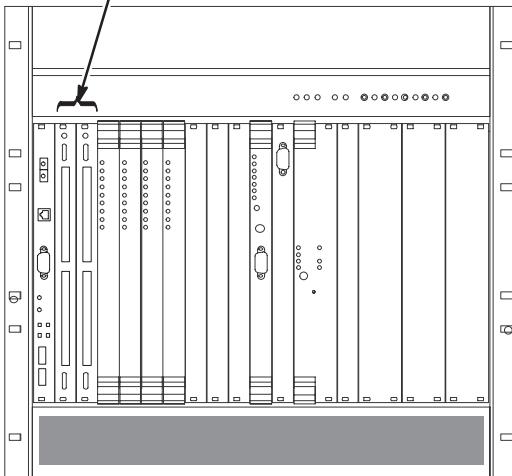
After upgrade, all systems will have one DRF2 board and IRF2 board.

Do NOT perform this procedure (Section 8-7) if site is not upgrading to 16 channels.

8-7-1 INSTALL REFLEX 800 AP BOARDS

Upgrade to a 16 Channel system requires the installation of ReFlex 800 AP boards, M3000RN option. Refer to Direction 5121980, also part of M3000RN, for instructions on installing AP boards.

- ① Remove blank panels or existing non-ReFlex 800 boards.
- ② Install new Master (2325947-15) and Slave (2325847-16). These boards are part of M3000RN.

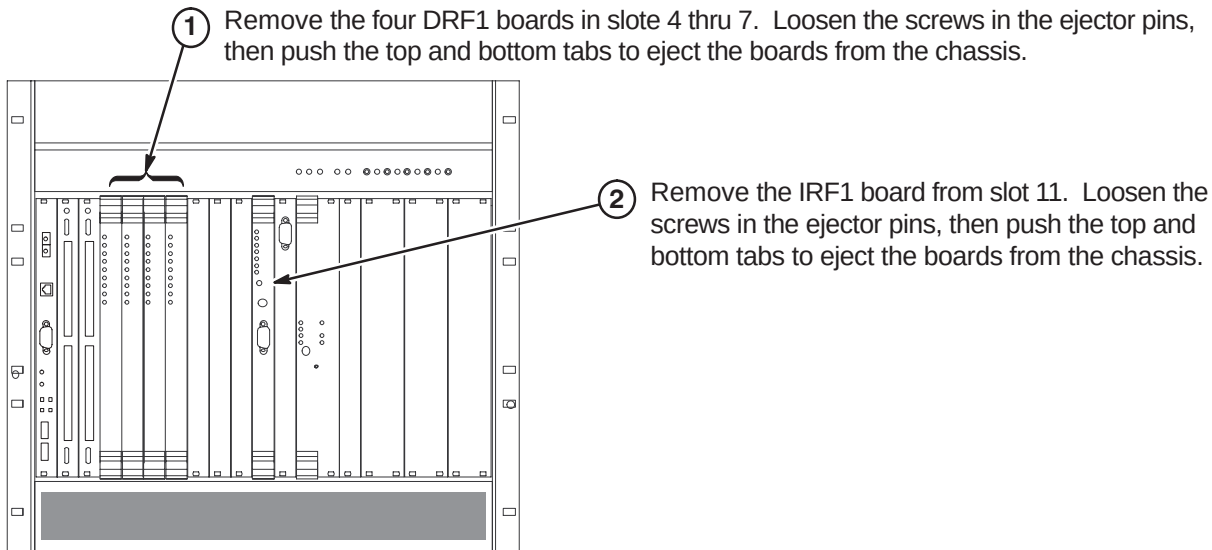


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

8-7-2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS



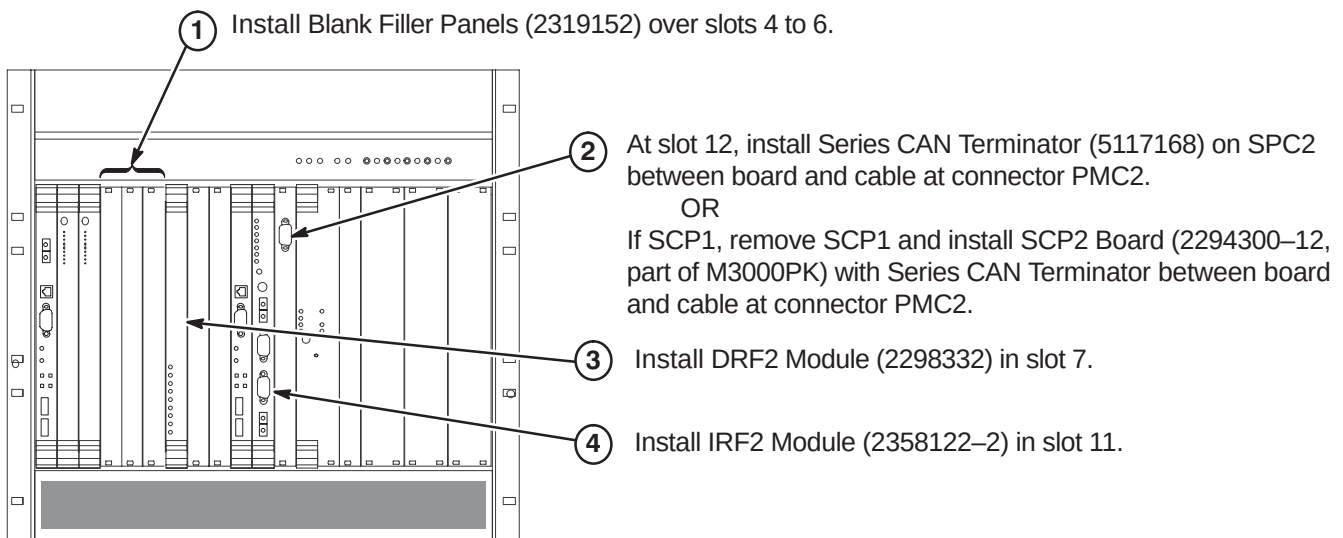
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



8-7-3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS

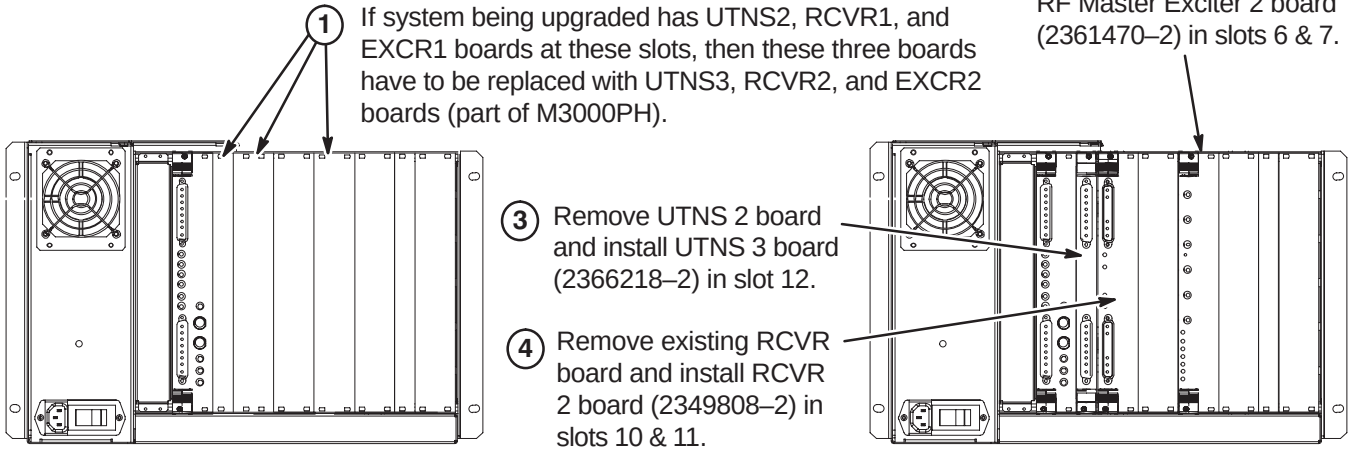


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



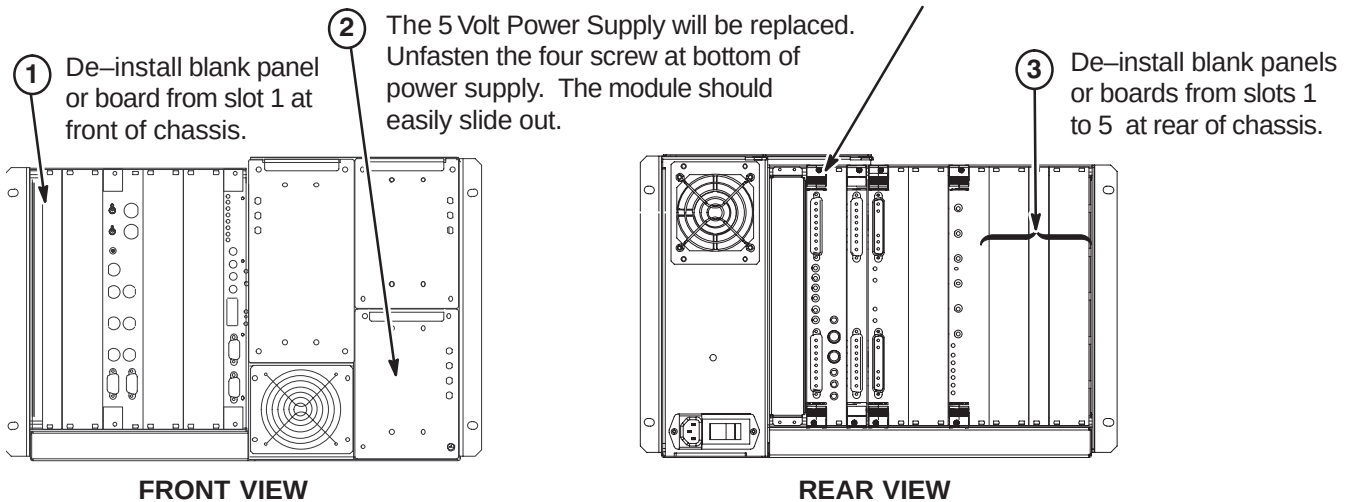
8-7-4 INSTALL UTNS3, RCVR2, AND EXCR2 BOARDS (If Required)

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

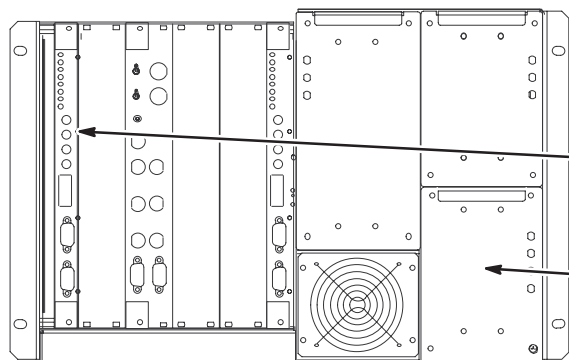
**8-7-5 REMOVE BLANK PANELS AND POWER SUPPLY FROM RRF CHASSIS (16 Channel Only)**

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.



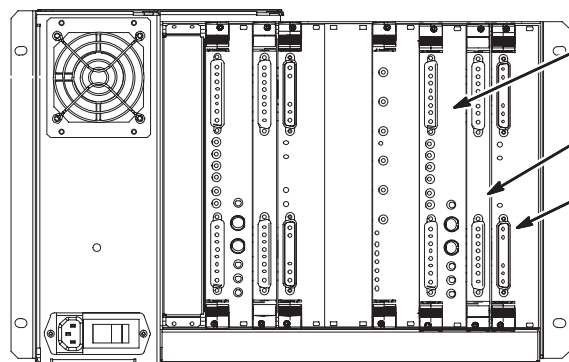
Note: All de-installed boards are to be returned for recycling.

8-7-6 INSTALL MODULES IN RRF CHASSIS (16 Channel Only)

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

① Install RF-DIF Board (2274184 or 5250028) in slot 1.

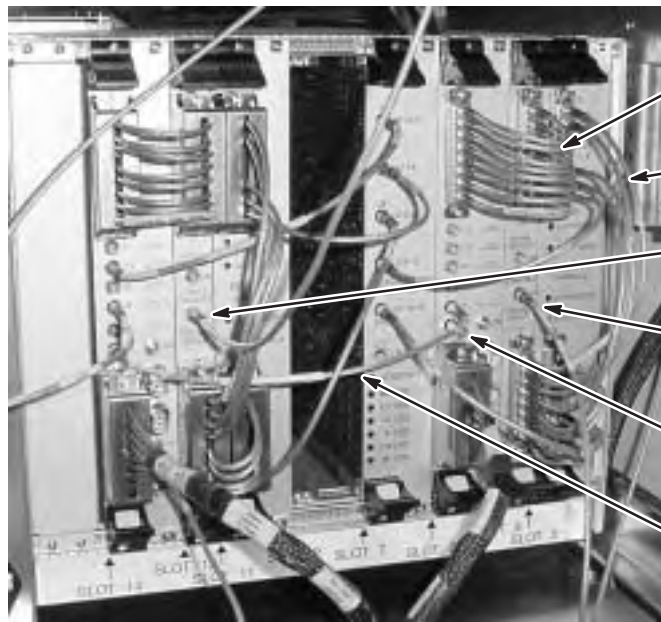
② Slide in new 5V RF Module RF Power Supply and secure with four screws previously removed.

FRONT

③ Install MUX Board (5250106) in slots 4 and 5.

④ Install UTNS3 Board (2366218-2) in slot 3.

⑤ Install RCVR II Board (2349808-2) in slots 1 and 2.

REAR**8-7-7 INSTALL CABLES AT REAR OF RRF CHASSIS**

① Connect cable 5123706 as marked.

② Connect cable 5113193-2 as marked.

③ Connect cable 5115417 slot 12R J31 as marked. Route and connect to Coax Power Splitter "1".

④ Connect cable 5115417-2 as marked. Route and connect to Coax Power Splitter "2".

⑤ Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).

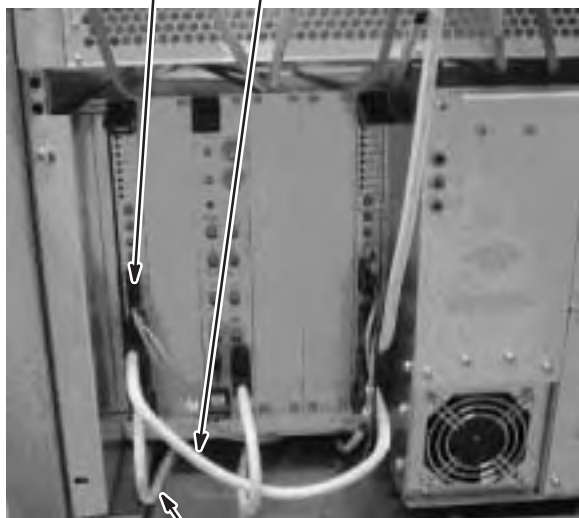
⑥ Connect cable 5115630 as marked.

8-7-8 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS

③ Connect cable 5121122-2 as marked to RF Slot 1F J21. Route and connect other end to MGD Slot 11F J7.

② Connect cable 5113182-2 as marked.

④ Slide in new 5V Power Supply (2292267-2). Make sure that it is properly seated before tightening the four screws.



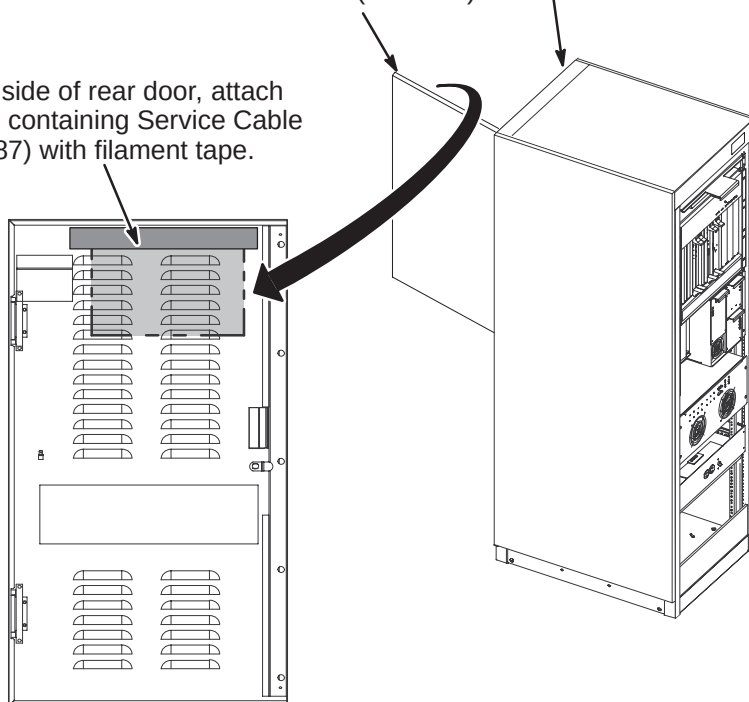
① Disconnect cable (2323878-3) from Slot 10F J23 and connect to Slot 1F J23.

**8-8 INSTALL NEW REAR DOOR AND RATING PLATE**

① Install new rear System Cabinet door (5115834).

③ Attach upgrade Rating Plate (46-302200P12) to top of cabinet on back side.

② At top of inside of rear door, attach plastic bag containing Service Cable Kit (5117087) with filament tape.



SECTION 9 – RF ACCESSORY CABINET

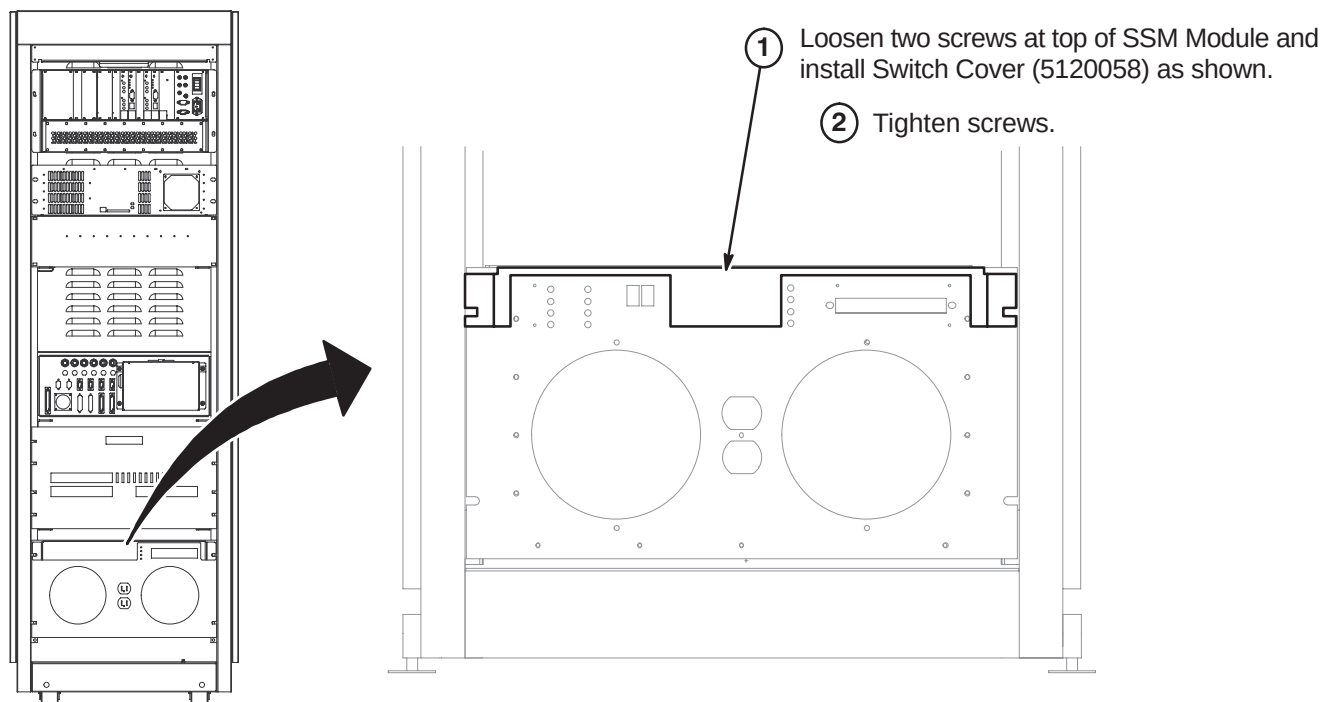
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

9-1 INTRODUCTION

The existing RF Accessory Cabinet will not be affected by the EXCITE upgrade to 12.x. Perform the following procedure only if cover is not already installed.

9-2 INSTALL SWITCH COVER



SECTION 10 – COMPLETION

10-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 10-2, 10-3, and 10-4.

10-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 10-3. Recycling Operation will route them to appropriate location.

10-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

10-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 11.x has been made before returning system to the users. Turn site over to applications who will instruct users.

10-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.